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Combustor size rating for a gas turbine engine using hydrogen fuel

Abstract

A gas turbine engine includes a hydrogen fuel delivery assembly configured to deliver a hydrogen fuel flow, a compressor section configured to compress air flowing therethrough to provide a compressed air flow, and a combustor including a combustion chamber having a burner length and a burner dome height. The combustion chamber is configured to combust a mixture of the hydrogen fuel flow and the compressed air flow. The combustion chamber can be characterized by a combustor size rating between one inch and seven inches. In more detail, the combustion chamber can be characterized by the combustor size rating between one inch and seven inches at a core air flow parameter between two and one half kN and sixty kN, in which the combustor size rating is a function of the core air flow parameter.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3917173	12/1974	Singh	N/A	N/A
4375150	12/1982	Nikiforakis	N/A	N/A
4794754	12/1988	Shekleton et al.	N/A	N/A
5109670	12/1991	Harshman	N/A	N/A
5307633	12/1993	Koemer et al.	N/A	N/A
5335502	12/1993	Roberts, Jr. et al.	N/A	N/A
5408830	12/1994	Lovett	N/A	N/A
5675971	12/1996	Angel et al.	N/A	N/A
5822992	12/1997	Dean	N/A	N/A
6415594	12/2001	Durbin et al.	N/A	N/A
6810673	12/2003	Snyder	N/A	N/A
6874323	12/2004	Stuttaford	N/A	N/A
7014835	12/2005	Mathias et al.	N/A	N/A
7900457	12/2010	Patterson et al.	N/A	N/A
8726626	12/2013	Spooner	N/A	N/A
9103547	12/2014	Eroglu et al.	N/A	N/A
9581085	12/2016	Bartz et al.	N/A	N/A
10101031	12/2017	Williams et al.	N/A	N/A
10317080	12/2018	Tu, Jr. et al.	N/A	N/A
10775047	12/2019	Horikawa et al.	N/A	N/A
11674443	12/2022	McCurdy Gibson et al.	N/A	N/A
11732652	12/2022	Sibbach et al.	N/A	N/A
12078100	12/2023	Benjamin	N/A	B64D 27/10
2005/0287407	12/2004	Bushko	N/A	N/A
2007/0271927	12/2006	Myers et al.	N/A	N/A
2008/0078182	12/2007	Evulet	N/A	N/A
2008/0163627	12/2007	ElKady et al.	N/A	N/A
2008/0256924	12/2007	Pederson et al.	N/A	N/A
2009/0113893	12/2008	Li et al.	N/A	N/A
2009/0199563	12/2008	Chen	N/A	N/A

2010/0101229	12/2009	York et al.	N/A	N/A
2010/0293959	12/2009	Remy et al.	N/A	N/A
2011/0314827	12/2010	Khosla et al.	N/A	N/A
2012/0079829	12/2011	Berry et al.	N/A	N/A
2012/0097757	12/2011	Bathina	N/A	N/A
2013/0086910	12/2012	Khan et al.	N/A	N/A
2013/0219899	12/2012	Uhm et al.	N/A	N/A
2014/0090392	12/2013	Meisner et al.	N/A	N/A
2014/0338338	12/2013	Chila et al.	N/A	N/A
2015/0323189	12/2014	Jeney et al.	N/A	N/A
2016/0363319	12/2015	Monahan et al.	N/A	N/A
2017/0122211	12/2016	Zhang et al.	N/A	N/A
2017/0298817	12/2016	Horiuchi et al.	N/A	N/A
2017/0343217	12/2016	Chen et al.	N/A	N/A
2018/0023812	12/2017	Laster	239/418	F23R 3/36
2018/0128489	12/2017	Boardman et al.	N/A	N/A
2018/0128490	12/2017	Boardman et al.	N/A	N/A
2018/0128491	12/2017	Boardman et al.	N/A	N/A
2018/0187603	12/2017	Berry	N/A	N/A
2019/0017441	12/2018	Venkatesan et al.	N/A	N/A
2019/0107053	12/2018	Miyamoto et al.	N/A	N/A
2020/0025385	12/2019	Boardman et al.	N/A	N/A
2020/0095956	12/2019	Ortelt et al.	N/A	N/A
2021/0010674	12/2020	Thariyan et al.	N/A	N/A
2021/0071590	12/2020	Beita et al.	N/A	N/A
2021/0071870	12/2020	Bulat	N/A	N/A
2021/0207808	12/2020	Mishra et al.	N/A	N/A
2021/0341147	12/2020	Wada et al.	N/A	N/A
2022/0030631	12/2021	Jung et al.	N/A	N/A
2022/0307428	12/2021	Sibbach et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	**Application Date**	**Country**	**CPC**
111553044	12/2019	CN	N/A
112231903	12/2020	CN	N/A
212537825	12/2020	CN	N/A
1736707	12/2017	EP	N/A
2462902	12/2009	GB	N/A
5965606	12/2011	JP	N/A
2012112642	12/2011	JP	N/A
2011086336	12/2010	WO	N/A

OTHER PUBLICATIONS

Lefebvre et al., “Gas Turbine Combustion: Alternative Fuels and Emissions,” CRC Press, 3rd Edition (Apr. 26, 2010). cited by applicant
Morgan et al., “Longitudinal Instability Limits With a Variable-Length Hydrogen-Oxygen Combustor,” NASA Technical Note, Lewis Research Center, National Aeronautics and Space Administration (1971). cited by applicant
Hua et al., “Numerical simulation of combustion of hydrogen air mixture in micro-scaled chamber. Part I: Fundamental Study” Mar. 31, 2005 (Year: 2005). cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) (1) This application is a continuation-in-part of U.S. application Ser. No. 17/457,559, filed Dec. 3, 2021, now allowed, which is incorporated herein by reference in its entirety.

TECHNICAL FIELD
(1) The present disclosure relates to a combustor for a gas turbine engine using hydrogen fuel, and in particular, for a gas turbine engine for aircraft.
BACKGROUND
(2) The propulsion system for commercial aircraft typically includes one or more aircraft engines, such as turbofan jet engines. The aircraft engine(s) may be mounted to a respective one of the wings of the aircraft, such as in a suspended position beneath the wing using a pylon. These engines may be powered by aviation turbine fuel, which is typically a combustible hydrocarbon liquid fuel, such as a kerosene-type fuel, having a desired carbon number and carbon to hydrogen ratio. Such fuel produces carbon dioxide emissions upon combustion and improvements to reduce such carbon dioxide emissions in commercial aircraft are desired.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) Features and advantages of the present disclosure will be apparent from the following, more particular, description of various exemplary embodiments, as illustrated in the accompanying drawings, wherein like reference numbers generally indicate identical, functionally similar, and/or structurally similar elements.
(2) FIG. **1** is a schematic perspective view of an aircraft having a gas turbine engine according to an embodiment of the present disclosure.
(3) FIG. **2** is a schematic, cross-sectional view, taken along line **2-2** in FIG. **1**, of the gas turbine engine of the aircraft shown in FIG. **1**.
(4) FIG. **3** is a perspective view of an unducted single fan engine that may be used with the aircraft shown in FIG. **1**.
(5) FIG. **4** is a schematic, cross-sectional view, taken along line **4-4** in FIG. **3**, of the unducted single fan engine shown in FIG. **3**.
(6) FIG. **5** is a schematic, cross-sectional view, taken along line **2-2** in FIG. **1**, of a turbojet engine that may be used with the aircraft shown in FIG. **1**.
(7) FIG. **6** is cross-sectional view of a first combustor for the gas turbine engine shown in FIG. **2**, showing detail **6** of FIG. **2**.
(8) FIG. **7** is cross-sectional view of a second combustor for the gas turbine engine shown in FIG. **2**, showing detail **6** of FIG. **2**.
(9) FIG. **8** is cross-sectional view of a third combustor for the gas turbine engine shown in FIG. **2**, showing detail **6** of FIG. **2**.
(10) FIG. **9** is cross-sectional view of a fourth combustor for the gas turbine engine shown in FIG. **2**, showing detail **6** of FIG. **2**.
(11) FIG. **10** is cross-sectional view of a fifth combustor for the gas turbine engine shown in FIG.

2, showing detail **6** of FIG. **2**.
(12) FIG. **11** is a graph illustrating combustor length (squared) as a function of combustor height, according to embodiments of the present disclosure.
(13) FIG. **12** is a graph illustrating combustor size rating as a function of core air flow parameter in engine gas turbine engines using hydrogen fuel, according to embodiments of the present disclosure.
(14) FIG. **13** is a schematic cross-sectional diagram of a turbine engine for an aircraft having a fuel-air mixing assembly in accordance with an exemplary embodiment of the present disclosure.
(15) FIG. **14** is a cross-sectional view of a portion of a combustor section of the turbine engine, further illustrating the fuel-air mixing assembly of FIG. **13** in accordance with an exemplary embodiment of the present disclosure.
(16) FIGS. **15**A-**15**E are exemplary cross sections of at least one fuel orifice of the fuel-air mixing assembly of FIG. **14** in accordance with the present disclosure.
(17) FIG. **16** is a variation of the cross-sectional view of FIG. **14** in accordance with an exemplary embodiment of the present disclosure.
(18) FIG. **17** is a cross-sectional view taken from FIG. **16**, further illustrating a first set of orifices in accordance with an exemplary embodiment of the present disclosure.
(19) FIG. **18** is cross-sectional view taken from FIG. **16**, further illustrating a second set of orifices in accordance with an exemplary embodiment of the present disclosure.
(20) FIG. **19** is a variation of the cross section of FIG. **16** in accordance with an exemplary embodiment of the present disclosure.
(21) FIG. **20** is another variation of the cross section of FIG. **16** in accordance with an exemplary embodiment of the present disclosure.
(22) FIG. **21** is yet another variation of the cross section of FIG. **14** in accordance with an exemplary embodiment of the present disclosure.
(23) FIG. **22** is cross section from FIG. **21** taken along a fuel orifice centerline in accordance with an exemplary embodiment of the present disclosure.
(24) FIG. **23** is a variation of the cross section of FIG. **22** in accordance with an exemplary embodiment of the present disclosure.
(25) FIG. **24** is another variation of the cross section of FIG. **22** in accordance with an exemplary embodiment of the present disclosure.
(26) FIG. **25** is still yet another variation of the cross section of FIG. **14** in accordance with an exemplary embodiment of the present disclosure.
(27) FIG. **26** is another variation of the cross section of FIG. **14** in accordance with an exemplary embodiment of the present disclosure.

DETAILED DESCRIPTION

(28) Features, advantages, and embodiments of the present disclosure are set forth or apparent from a consideration of the following detailed description, drawings, and claims. Moreover, it is to be understood that the following detailed description is exemplary and intended to provide further explanation without limiting the scope of the disclosure as claimed.
(29) Various embodiments are discussed in detail below. While specific embodiments are discussed, this is done for illustration purposes only. A person skilled in the relevant art will recognize that other components and configurations may be used without departing from the spirit and scope of the present disclosure.
(30) As used herein, the terms “first,” “second,” and “third” may be used interchangeably to distinguish one component from another and are not intended to signify location or importance of the individual components.
(31) The terms “forward” and “aft” refer to relative positions within a gas turbine engine or vehicle, and refer to the normal operational attitude of the gas turbine engine or vehicle. For example, with regard to a gas turbine engine, forward refers to a position closer to an engine inlet

and aft refers to a position closer to an engine nozzle or exhaust.
(32) The terms “upstream” and “downstream” refer to the relative direction with respect to fluid flow in a fluid pathway. For example, “upstream” refers to the direction from which the fluid flows, and “downstream” refers to the direction to which the fluid flows.
(33) The terms “coupled,” “fixed,” “attached,” “connected,” and the like, refer to both direct coupling, fixing, attaching, or connecting as well as indirect coupling, fixing, attaching, or connecting through one or more intermediate components or features, unless otherwise specified herein.
(34) The singular forms “a,” “an,” and “the” include plural references unless the context clearly dictates otherwise.
(35) Approximating language, as used herein throughout the specification and claims, is applied to modify any quantitative representation that could permissibly vary without resulting in a change in the basic function to which it is related. Accordingly, a value modified by a term or terms, such as “about,” “approximately,” and “substantially” are not to be limited to the precise value specified. In at least some instances, the approximating language may correspond to the precision of an instrument for measuring the value, or the precision of the methods or machines for constructing or manufacturing the components and/or systems. For example, the approximating language may refer to being within a one, two, four, ten, fifteen, or twenty percent margin in either individual values, range(s) of values and/or endpoints defining range(s) of values.
(36) The term “bypass ratio,” unless stated otherwise, means the bypass ratio at take off conditions. The term bypass ratio as used herein means the ratio between the mass flow rate of air flow accelerated by the engine that bypasses the engine core to the mass flow rate of the air flow entering the engine core. For example, in an exemplary engine such as the turbofan engine **100** depicted in FIG. **2** and discussed further below, the bypass ratio is the ratio of the mass flow rate of the air flow entering the bypass air flow passage **140** to the mass flow rate of the air flow entering the core air flow path **121**. The bypass ratio can also be estimated as a ratio of the area of an inlet to the bypass duct (e.g., inlet of the bypass air flow passage **140**, discussed below) or an area swept by a rotor (e.g., the area swept by fan blades **322**, discussed below) to the area of the inlet to the engine core (e.g., inlet of the core air flow path **121**).
(37) The term “thrust,” unless stated otherwise, means the maximum thrust at take off. This meaning of thrust is adopted when computing a core airflow parameter (relationship (2), below).
(38) Here and throughout the specification and claims, range limitations are combined, and interchanged. Such ranges are identified and include all the sub-ranges contained therein unless context or language indicates otherwise. For example, all ranges disclosed herein are inclusive of the endpoints, and the endpoints are independently combinable with each other.
(39) Combustible hydrocarbon liquid fuel, such as Jet-A fuel, has long been used in gas turbine engines and the components of gas turbine engines, particularly, the combustor, have been designed for such fuels. A hydrogen fuel may be utilized to eliminate carbon dioxide emissions from commercial aircraft. Hydrogen fuel, however, poses a number of challenges as compared to combustible hydrocarbon liquid fuel, such as Jet-A fuel. Hydrogen fuel, for example, is a highly reactive fuel that burns at higher temperatures than combustible hydrocarbon liquid fuel. Hydrogen fuel also has much higher flame speeds. For example, the laminar flame speed for a hydrogen fuel of diatomic hydrogen is an order of magnitude greater than the laminar flame speed for Jet-A fuel.
(40) When testing hydrogen fuel in current gas turbine engines with rich burn combustors, we, the inventors, observed that the higher combustion temperature of hydrogen fuel results in increased production of nitrogen oxides (“NOx”), as compared to combustible hydrocarbon liquid fuel. We also observed in our testing that NOx emissions are sensitive to combustor residence time. As noted above, hydrogen fuel is highly reactive (relative to other fuels) with a wide range of flammability limits and very high flame speeds, resulting in a very short hydrogen flame close to the front end of the combustor. With such a short flame, the post-flame residence time increases for

combustors designed for Jet-A fuel. These findings resulted in a realization that when designing a hydrogen fuel combustor to meet NOx emission targets, the combustor residence time needs to be reduced by more than about fifty percent. To find a suitable combustor design for gas turbine engines using hydrogen fuel, we conceived of a wide variety of combustors having different shapes and sizes in order to determine which embodiment(s) were most promising for a variety of contemplated engine designs and thrust classes. The various embodiments, as described herein and as shown in the figures, are combustors that are sized to meet NOx emissions targets.
(41) FIG. **1** is a perspective view of an aircraft **10** that may implement various preferred embodiments. The aircraft **10** includes a fuselage **12**, wings **14** attached to the fuselage **12**, and an empennage **16**. The aircraft **10** also includes a propulsion system that produces a propulsive thrust required to propel the aircraft **10** in flight, during taxiing operations, and the like. The propulsion system for the aircraft **10** shown in FIG. **1** includes a pair of engines **20**. In this embodiment, each engine **20** is attached to one of the wings **14** by a pylon **18** in an under-wing configuration. Although the engines **20** are shown attached to the wing **14** in an under-wing configuration in FIG. **1**, in other embodiments, the engine **20** may have alternative configurations and be coupled to other portions of the aircraft **10**. For example, the engine **20** may additionally or alternatively include one or more aspects coupled to other parts of the aircraft **10**, such as, for example, the empennage **16** and the fuselage **12**.
(42) As will be described further below with reference to FIG. **2**, the engines **20** shown in FIG. **1** are gas turbine engines that are each capable of selectively generating a propulsive thrust for the aircraft **10**. The amount of propulsive thrust may be controlled at least in part based on a volume of fuel provided to the gas turbine engines **20** via a fuel system **200**. The fuel is stored in a fuel tank **212** of the fuel system **200**. As shown in FIG. **1**, at least a portion of the fuel tank **212** is located in each wing **14** and a portion of the fuel tank **212** is located in the fuselage **12** between the wings **14**. The fuel tank **212**, however, may be located at other suitable locations in the fuselage **12** or the wing **14**. The fuel tank **212** may also be located entirely within the fuselage **12** or the wing **14**. The fuel tank **212** may also be separate tanks instead of a single, unitary body, such as, for example, two tanks each located within a corresponding wing **14**.
(43) Although the aircraft **10** shown in FIG. **1** is an airplane, the embodiments described herein may also be applicable to other aircraft **10**, including, for example, helicopters and unmanned aerial vehicles (UAV). The aircraft discussed herein are fixed-wing aircraft or rotor aircraft that generate lift by aerodynamic forces acting on, for example, a fixed wing (e.g., wing **14**) or a rotary wing (e.g., rotor of a helicopter), and are heavier-than-air aircraft, as opposed to lighter-than-air aircraft (such as a dirigible). The engine **20** may be used in various other applications including stationary power generation systems and other vehicles beyond the aircraft **10** explicitly described herein, such as boats, ships, cars, trucks, and the like.
(44) FIG. **2** is a schematic, cross-sectional view of one of the engines **20** used in the propulsion system for the aircraft **10** shown in FIG. **1**. The cross-sectional view of FIG. **2** is taken along line **2-2** in FIG. **1**. For the embodiment depicted in FIG. **2**, the engine **20** is a high bypass turbofan engine that is referred to as a turbofan engine **100** herein. The turbofan engine **100** has an axial direction A (extending parallel to a longitudinal centerline axis **101**, shown for reference in FIG. **2**), a radial direction R, and a circumferential direction. The circumferential direction (not depicted in FIG. **2**) extends in a direction rotating about the axial direction A. The turbofan engine **100** includes a fan section **102** and a turbomachine **104** disposed downstream from the fan section **102**.
(45) The turbomachine **104** depicted in FIG. **2** includes a tubular outer housing or nacelle **106** and an inlet **108**. Within the housing **106** there is an engine core, which includes, in a serial flow relationship, a compressor section including a booster or low-pressure (LP) compressor **110** and a high-pressure (HP) compressor **112**, a combustion section **150** (also referred to herein as a combustor **150**), a turbine section including a high-pressure (HP) turbine **116** and a low-pressure (LP) turbine **118**, and a jet exhaust nozzle section **120**. The compressor section, the combustor **150**,

and the turbine section together define at least in part a core air flow path **121** extending from the inlet **108** to the jet exhaust nozzle section **120**. The turbofan engine **100** further includes one or more drive shafts. More specifically, the turbofan engine includes a high-pressure (HP) shaft or spool **122** drivingly connecting the HP turbine **116** to the HP compressor **112**, and a low-pressure (LP) shaft or spool **124** drivingly connecting the LP turbine **118** to the LP compressor **110**.
(46) The fan section **102** shown in FIG. **2** includes a fan **126** having a plurality of fan blades **128** coupled to a disk **130**. The fan blades **128** and the disk **130** are rotatable, together, about the longitudinal centerline axis **101** by the LP shaft **124**. The booster **108** may also be directly driven by the LP shaft **124**, as depicted in FIG. **2**. The disk **130** is covered by a rotatable front hub **132** aerodynamically contoured to promote an air flow through the plurality of fan blades **128**. Further, an annular fan casing or outer nacelle **134** is provided, circumferentially surrounding the fan **126** and/or at least a portion of the turbomachine **104**. The nacelle **134** is supported relative to the turbomachine **104** by a plurality of circumferentially spaced outlet guide vanes **136**. A downstream section **138** of the nacelle **134** extends over an outer portion of the turbomachine **104** so as to define a bypass air flow passage **140** therebetween.
(47) The turbofan engine **100** is operable with the fuel system **200** and receives a flow of fuel from the fuel system **200**. As will be described further below, the fuel system **200** includes a fuel delivery assembly **202** providing the fuel flow from the fuel tank **212** to the turbofan engine **100**, and, more specifically, to a plurality of fuel nozzles **442** that inject fuel into a combustion chamber **430** of the combustor **150**.
(48) The turbofan engine **100** also includes various accessory systems to aid in the operation of the turbofan engine **100** and/or an aircraft including the turbofan engine **100**. For example, the turbofan engine **100** may include a main lubrication system **171**, a compressor cooling air (CCA) system **173**, an active thermal clearance control (ATCC) system **175**, and a generator lubrication system **177**, each of which is depicted schematically in FIG. **2**. The main lubrication system **171** is configured to provide a lubricant to, for example, various bearings and gear meshes in the compressor section, the turbine section, the HP spool **122**, and the LP shaft **124**. The lubricant provided by the main lubrication system **171** may increase the useful life of such components and may remove a certain amount of heat from such components through the use of one or more heat exchangers. The compressor cooling air (CCA) system **173** provides air from one or both of the HP compressor **112** or LP compressor **110** to one or both of the HP turbine **116** or LP turbine **118**. The active thermal clearance control (ATCC) system **175** acts to minimize a clearance between tips of turbine blades and casing walls as casing temperatures vary during a flight mission. The generator lubrication system **177** provides lubrication to an electronic generator (not shown), as well as cooling/heat removal for the electronic generator. The electronic generator may provide electrical power to, for example, a startup electrical motor for the turbofan engine **100** and/or various other electronic components of the turbofan engine **100** and/or an aircraft including the turbofan engine **100**.
(49) Heat from these accessory systems **171**, **173**, **175**, and **177**, and other accessory systems, may be provided to various heat sinks as waste heat from the turbofan engine **100** during operation, such as to various vaporizers **220**, as discussed below. Additionally, the turbofan engine **100** may include one or more heat exchangers **179** within, for example, the turbine section or jet exhaust nozzle section **120** for extracting waste heat from an air flow therethrough to also provide heat to various heat sinks, such as the vaporizers **220**, discussed below.
(50) The fuel system **200** of this embodiment is configured to store the fuel for the turbofan engine **100** in the fuel tank **212** and to deliver the fuel to the turbofan engine **100** via the fuel delivery assembly **202**. The fuel delivery assembly **202** includes tubes, pipes, and the like, to fluidly connect the various components of the fuel system **200** to the turbofan engine **100**. As discussed above, the turbofan engine **100**, and, in particular, the combustor **150** discussed herein may be particularly suited for use with hydrogen fuel (diatomic hydrogen). In the embodiments shown in FIG. **2**, the

fuel is a hydrogen fuel comprising hydrogen, more specifically, diatomic hydrogen. In some embodiments, the hydrogen fuel may consist essentially of hydrogen.

(51) The fuel tank **212** may be configured to hold the hydrogen fuel at least partially in the liquid phase and may be configured to provide hydrogen fuel to the fuel delivery assembly **202** substantially completely in the liquid phase, such as completely in the liquid phase. For example, the fuel tank **212** may have a fixed volume and contain a volume of the hydrogen fuel in the liquid phase (liquid hydrogen fuel). As the fuel tank **212** provides hydrogen fuel to the fuel delivery assembly **202** substantially completely in the liquid phase, the volume of the liquid hydrogen fuel in the fuel tank **212** decreases and the remaining volume in the fuel tank **212** is made up by, for example, hydrogen in the gaseous phase (gaseous hydrogen). As used herein, the term “substantially completely” as used to describe a phase of the hydrogen fuel, refers to at least 99% by mass of the described portion of the hydrogen fuel being in the stated phase, such as at least 97.5%, such as at least 95%, such as at least 92.5%, such as at least 90%, such as at least 85%, or such as at least 75% by mass of the described portion of the hydrogen fuel being in the stated phase.

(52) To store the hydrogen fuel substantially completely in the liquid phase, the hydrogen fuel is stored in the fuel tank **212** at very low (cryogenic) temperatures. For example, the hydrogen fuel may be stored in the fuel tank **212** at about-253 degrees Celsius or less at atmospheric pressure, or at other temperatures and pressures to maintain the hydrogen fuel substantially in the liquid phase. The fuel tank **212** may be made from known materials such as titanium, Inconel®, aluminum, or composite materials. The fuel tank **212** and the fuel system **200** may include a variety of supporting structures and components to facilitate storing the hydrogen fuel in such a manner.

(53) The liquid hydrogen fuel is supplied from the fuel tank **212** to the fuel delivery assembly **202**. The fuel delivery assembly **202** may include one or more lines, conduits, etc., configured to carry the hydrogen fuel between the fuel tank **212** and the turbofan engine **100**. The fuel delivery assembly **202** thus provides a flow path of the hydrogen fuel from the fuel tank **212** to the turbofan engine **100**. The hydrogen fuel is delivered to the engine by the fuel delivery assembly **202** in the gaseous phase, the supercritical phase, or both (e.g., the gaseous phase and the supercritical phase). The fuel system **200** thus includes a vaporizer **220** in fluid communication with the fuel delivery assembly **202** to heat the liquid hydrogen fuel flowing through the fuel delivery assembly **202**. The vaporizer **220** is positioned in the flow path of the hydrogen fuel between the fuel tank **212** and the turbofan engine **100**. The vaporizer **220** may be positioned at least partially within the fuselage **12** or the wing **14** (both shown in FIG. **1**), such as at least partially within the wing **14**. The vaporizer **220** may, however, be positioned at other suitable locations in the flow path of the hydrogen between the fuel tank **212** and the turbofan engine **100**. For example, the vaporizer **220** may be positioned external to the fuselage **12** and the wing **14** (both shown in FIG. **1**) and positioned at least partially within the pylon **18** (FIG. **1**) or the turbofan engine **100** (FIG. **2**). When positioned in the turbofan engine **100**, the vaporizer may be located in the nacelle **134**, for example. Although only one vaporizer **220** is shown in FIG. **2**, the fuel system **200** may include multiple vaporizers **220**. For example, when a vaporizer **220** is positioned in the turbofan engine **100** or in the pylon **18** and functions as a primary vaporizer configured to operate once the turbofan engine **100** is in a thermally stable condition, another vaporizer **220** is positioned upstream of the primary vaporizer and proximate to the fuel tank **212**, and functions as a primer vaporizer during start-up (or prior to start-up) of the turbofan engine **100**.

(54) The vaporizer **220** is in thermal communication with at least one heat source **222**, **224**. In this embodiment, the vaporizer **220** is in thermal communication with a primary heat source **222** and an auxiliary heat source **224**. In this embodiment, primary heat source **222** is waste heat from the turbofan engine **100**, and the vaporizer **220** is, thus, thermally connected to at least one of the main lubrication system **171**, the compressor cooling air (CCA) system **173**, the active thermal clearance control (ATCC) system **175**, the generator lubrication system **177**, and the heat exchangers **179** to

extract waste heat from the turbofan engine **100** to heat the hydrogen fuel. In such a manner, the vaporizer **220** is configured to operate by drawing heat from the primary heat source **222** once the turbofan engine **100** is capable of providing enough heat, via the auxiliary heat source **224**, to the vaporizer **220**, in order to facilitate operation of the vaporizer **220**.

(55) The vaporizer **220** may be heated by any suitable heat source, and, in this embodiment, for example, the auxiliary heat source **224** is a heat source external to the turbofan engine **100**. The auxiliary heat source **224** may include, for example, an electrical power source, a catalytic heater or burner, and/or a bleed air flow from an auxiliary power unit. The auxiliary heat source **224** may be integral to the vaporizer **220**, such as when the vaporizer **220** includes one or more electrical resistance heaters, or the like, that are powered by the electrical power source. In this configuration the auxiliary heat source **224** may provide heat for the vaporizer **220** independent of whether or not the turbofan engine **100** is running and can be used, for example, during start-up (or prior to start-up) of the turbofan engine **100**.

(56) As noted, the vaporizer **220** is in communication with the flow of the hydrogen fuel through the fuel delivery assembly **202**. The vaporizer **220** is configured to draw heat from at least one of the primary heat source **222** and the auxiliary heat source **224** to heat the flow of hydrogen fuel from a substantially completely liquid phase to a substantially completely gaseous phase or to a substantially completely supercritical phase.

(57) The fuel system **200** also includes a high-pressure pump **232** in fluid communication with the fuel delivery assembly **202** to induce the flow of the hydrogen fuel through the fuel delivery assembly **202** to the turbofan engine **100**. The high-pressure pump **232** may generally be the primary source of pressure rise in the fuel delivery assembly **202** between the fuel tank **212** and the turbofan engine **100**. The high-pressure pump **232** may be configured to increase a pressure in the fuel delivery assembly **202** to a pressure greater than a pressure within the combustion chamber **430** of the combustor **150** of the turbofan engine **100**, and to overcome any pressure drop of the components placed downstream of the high-pressure pump **232**.

(58) The high-pressure pump **232** is positioned within the flow of hydrogen fuel in the fuel delivery assembly **202** at a location downstream of the vaporizer **220**. In this embodiment, the high-pressure pump **232** is positioned external to the fuselage **12** and the wing **14**, and is positioned at least partially within the pylon **18**, or at least partially within the turbofan engine **100**. More specifically, the high-pressure pump **232** is positioned within the turbofan engine **100**. With the high-pressure pump **232** located in such a position, the high-pressure pump **232** may be any suitable pump configured to receive the flow of hydrogen fuel in substantially completely a gaseous phase or a supercritical phase. In other embodiments, however, the high-pressure pump **232** may be positioned at other suitable locations, including other positions within the flow path of the hydrogen fuel. For example, the high-pressure pump **232** may be located upstream of the vaporizer **220** and may be configured to receive the flow of hydrogen fuel through the fuel delivery assembly **202** in a substantially completely liquid phase.

(59) The fuel system **200** also includes a metering unit in fluid communication with the fuel delivery assembly **202**. Any suitable metering unit may be used including, for example, a fuel metering valve **234** placed in fluid communication with the fuel delivery assembly **202**. The fuel delivery assembly **202** is configured to provide the fuel metering valve **234**, and the fuel metering valve **234** is configured to receive hydrogen fuel. In this embodiment, the fuel metering valve **234** is positioned downstream of the high-pressure pump **232**. The fuel metering valve **234** is further configured to provide the flow of the hydrogen fuel to the turbofan engine **100** in a desired manner. The fuel metering valve **234** is configured to provide a desired volume of the fuel at, for example, a desired flow rate, to a fuel manifold **236** of the turbofan engine **100**. The fuel manifold **236** then distributes (provides) the hydrogen fuel received to a plurality of fuel nozzles **442** (see FIG. **6**) within the combustion section **150** of the turbofan engine **100** where the hydrogen fuel is mixed with compressed air, and the mixture of hydrogen fuel and compressed air is combusted to generate

combustion gases that drive the turbofan engine **100**. Adjusting the fuel metering valve **234** changes the volume of fuel provided to the combustion chamber **430** (see FIG. **6**) of the combustor **150** and, thus, changes the amount of propulsive thrust produced by the turbofan engine **100** to propel the aircraft **10**.

(60) Although the turbofan engine **100** is shown as a direct drive, fixed-pitch turbofan engine **100**, in other embodiments, a gas turbine engine may be a geared gas turbine engine (i.e., including a gearbox between the fan **126** and shaft driving the fan, such as the LP shaft **124**), may be a variable pitch gas turbine engine (i.e., including a fan **126** having a plurality of fan blades **128** rotatable about their respective pitch axes), etc. Additionally, in still other exemplary embodiments, the exemplary turbofan engine **100** may include or be operably connected to any other suitable accessory systems. Additionally, or alternatively, the exemplary turbofan engine **100** may not include or be operably connected to one or more of the accessory systems **171**, **173**, **175**, and **177**, discussed above.

(61) The turbofan engine **100** discussed herein is an example of the engine **20** in which the combustors **150** discussed herein may be used. In other embodiments, other suitable engines may be utilized with aspects of the present disclosure. For example, FIGS. **3** and **4** show an unducted single fan (USF) engine **300** that may be used as the engine **20** of the aircraft **10** and implement the fuel system described above, and combustor designs discussed further below. FIG. **3** is a perspective view of the USF engine **300** and FIG. **4** is a cross-sectional view taken along a line **4-4** in FIG. **3**.

(62) The USF engine **300** includes a housing **302**. The housing **302** may be formed of a nacelle **310** and spinner **320**. The nacelle **310** and/or the spinner **320** house internal components of the USF engine **300**. For example, the nacelle **310** houses a torque producing system **312** coupled to a shaft **314**. The torque producing system **312** in the embodiments discussed herein is a gas turbine engine, such as the turbomachine **104** discussed above with reference to FIG. **2** and, thus, the nacelle **310** of this embodiment is similar to the tubular outer housing **106** discussed above. As the turbomachine **104** used as the torque producing system **312** of the USF engine has the same or similar components and features as the turbomachine **104** discussed above, a detailed description of the components of the turbomachine **104** used in of the USF engine **300** is omitted.

(63) The torque producing system **312** and the shaft **314** are configured to operate (e.g., to rotate) the spinner **320**. One or more fan blades **322** are coupled to the spinner **320**. More specifically, the spinner **320** includes a fan hub **324**, and the fan blades **322** are coupled to the fan hub **324**. The spinner **320** rotates with respect to the nacelle **310**. Coupled to the nacelle **310** may be one or more outlet guide vanes **326**. In this embodiment, the outlet guide vanes **326** are positioned aft of the fan blades **322**. During operation, the one or more fan blades **322** (by virtue of the connection to the spinner **320**) rotate circumferentially around a longitudinal centerline **304**, in this embodiment, and the nacelle **310** is stationary such that the one or more outlet guide vanes **326** do not rotate around the longitudinal centerline **304** and are, thus, stationary with respect to rotation about the longitudinal centerline **304**. Although the outlet guide vanes **326** are stationary with respect to the longitudinal centerline **304**, the outlet guide vanes **326** are capable of being rotated or moved with respect to the nacelle **310**, for example, in the direction A of FIG. **4**.

(64) During operation of the USF engine **300**, air flows from the left side of FIG. **4** toward the right side of FIG. **4**. A portion of the air flow may flow past the fan blades **322** and the outlet guide vanes **326**. A portion of the air flow may enter the nacelle **310** through the annular inlet **108** to be mixed with the hydrogen fuel for combustion in a combustor **150** of the USF engine **300** and exit through an outlet **120**. The outlet guide vanes **326** may be movable with respect to the nacelle **310** to guide the air flow in a particular direction. Each outlet guide vane **326** may be movable to adjust the lean, pitch, sweep, or any combination thereof, of the outlet guide vane **326**.

(65) In the embodiment shown in FIGS. **3** and **4**, a forward end or front portion of the housing **302** includes the one or more fan blades **322** and the one or more outlet guide vanes **326**. In other

embodiments, the one or more fan blades **322** and the one or more outlet guide vanes **326** may have a different arrangement with respect to the housing **302**. For example, the one or more fan blades **322** and the one or more outlet guide vanes **326** may be located on an aft end or rear portion of the housing **302**, such as coupled to a rear portion of the housing **302**.

(66) In other embodiments, an engine according to the disclosure may be configured to have either the stationary vanes positioned forward of the rotating blades **322** (thus, the blades **326** are inlet guide vanes) or both the blades **326** and blades **322** configured to operate in a counter-rotating fashion. Either "pusher" or "puller" configurations are contemplated. In each of these alternative embodiments, the fuel delivery system **200** and combustor **150**, as described in great detail below, may be used. An example of a suitable engine configuration for a counter-rotating engine is shown and described in FIG. **1** and col. 3, line 43 through col. 4, line 11 of U.S. Pat. No. 10,800,512, hereby incorporated by reference for all purposes. Alternative embodiments of the USF engine **300** are shown and described in FIGS. **6**, **7**, and **8** and col. 4, line 51 through col. 5, line 19 of U.S. Pat. No. 10,704,410, hereby incorporated by reference for all purposes.

(67) In further embodiments, a turbojet engine **350** may be used as the engine **20**. FIG. **5** is a schematic, cross-sectional view of the turbojet engine **350**. The cross-sectional view of FIG. **5** is similar to FIG. **2**, which is taken along line **2-2** in FIG. **1**. The turbojet engine **350** includes the same or similar components of the turbomachine **104** of the turbofan engine **100** and a detailed description of these components is omitted. An exemplary turbojet engine **350** may not include a fan with bypass duct. An exemplary turbojet engine **350** may have high velocity exhaust from the engine, which produces a majority of the thrust for the turbojet engine **350**. In still further embodiments, other suitable gas turbine engines, such as a turboshaft engine, a turboprop engine, and the like, may be utilized with aspects of the present disclosure.

(68) As noted above, we conceived of a wide variety of combustors having different shapes and sizes. FIGS. **6** to **10** show various combustor shapes that can suitably be used as the combustor **150** for the gas turbine engines **20** discussed herein. FIGS. **6** to **10** are a detail views showing detail **6** in FIG. **2**, and, as FIG. **2** is a cross-sectional view, FIGS. **6** to **10** are also cross-sectional views. FIG. **6** shows a first combustor **401**. FIG. **7** shows a second combustor **403**. FIG. **8** shows a third combustor **405**. FIG. **9** shows a fourth combustor **407**. FIG. **10** shows a fifth combustor **409**. Although the shapes of these combustors **401**, **403**, **405**, **407**, **409** differ, each of these combustors **401**, **403**, **405**, **407**, **409** has similar components, and common reference numerals are used in FIGS. **6** to **10** to for the same or similar components of these combustors **401**, **403**, **405**, **407**, **409**. Accordingly, the following detailed description of the first combustor **401** also applies to the second combustor **403**, the third combustor **405**, the fourth combustor **407**, and the fifth combustor **409**. Some components, such as the combustor casing **410**, for example, may not be shown in each figure, but such components may nevertheless be applicable to the combustors **403**, **405**, **407**, **409**.

(69) As shown in FIG. **6**, combustor **401** includes a combustor casing **410** and a combustor liner **420**. The combustor casing **410** of this embodiment has an outer casing **412** and an inner casing **414**, and the combustor liner **420** of this embodiment has an outer liner **422** and an inner liner **424**. A combustion chamber **430** is formed within the combustor liner **420**. More specifically, the outer liner **422** and the inner liner **424** are disposed between the outer casing **412** and the inner casing **414**. The outer liner **422** and the inner liner **424** are spaced radially from each other such that the combustion chamber **430** is defined therebetween. The outer casing **412** and the outer liner **422** form an outer passage **416** therebetween, and the inner casing **414** and the inner liner **424** form an inner passage **418** therebetween. In this embodiment, the combustor **401** is a single annular combustor, but, in other embodiments, the combustor **401** may be any other combustor, including, but not limited to a double annular combustor.

(70) The combustion chamber **430** has a forward end **432** (downstream end) and an aft end **434** (upstream end). The fuel nozzle **442** is positioned at the forward end **432** of the combustion chamber **430**. The fuel nozzle **442** of this embodiment is part of a swirler/fuel nozzle assembly **440**.

In this embodiment, when the combustor **401** is an annular combustor **150**, a plurality of fuel nozzles **442** is arranged in an annular configuration with the plurality of fuel nozzles **442** (the swirler/fuel nozzle assemblies **440**) aligned in a circumferential direction of the combustor **401**.
(71) As discussed above, the compressor section, the combustor **401**, and the turbine section form, at least in part, the core air flow path **121** extending from the annular inlet **108** to the jet exhaust nozzle section **120**. Air entering through the annular inlet **108** is compressed by blades of a plurality of fans of the LP compressor **110** and HP compressor **112**. A cowl assembly **450** is coupled to the upstream ends of outer liner **422** and the inner liner **424**, respectively. An annular opening **452** formed in the cowl assembly **450** enables compressed air from the compressor section (indicated by arrow B) to enter the combustor **401**. The compressed air flows through the annular opening **452** to support combustion. Another portion of the compressed air flows around the outside of the combustor liner **420** through the outer passage **416** and the inner passage **418**. This air is introduced into the combustion chamber **430** through a plurality of circumferentially spaced dilution holes **426** formed in the combustor liner **420** at positions downstream of the fuel nozzle **442**.
(72) An annular dome plate **454** extends between, and is coupled to, outer liner **422** and the inner liner **424** near their upstream ends. The plurality of circumferentially spaced swirler/fuel nozzle assemblies **440** is coupled to dome plate **454**. Each swirler/fuel nozzle assembly **440** receives compressed air from the annular opening **452**. The swirler/fuel nozzle assembly **440** includes a swirler **444** that is used to generate turbulence in the air. The fuel nozzle **442** injects fuel into the turbulent air flow and the turbulence promotes rapid mixing of the fuel with the air. The resulting mixture of fuel and compressed air is discharged into combustion chamber **430** and combusted in the combustion chamber **430**, generating combustion gases (combustion products), which accelerate as the combustion gases leave the combustion chamber **430**.
(73) A turbine nozzle **460** is disposed at the outlet of the combustion chamber **430**. The turbine nozzle **460** may be a stage **1** turbine nozzle. The turbine nozzle **460** is coupled to outer liner **422** and the inner liner **424** at the downstream (aft) ends of each of the outer liner **422** and the inner liner **424**. The turbine nozzle **460** of this embodiment includes an outer band **462** and an inner band **464** coupled to outer liner **422** and the inner liner **424**, respectively. The turbine nozzle **460** also includes a leading edge **466**, which in this embodiment is the location where the turbine nozzle **460** is coupled to outer liner **422** and the inner liner **424**, and the outer band **462** and the inner band **464** each has the leading edge **466**. The turbine nozzle **460** further includes a plurality of circumferentially spaced vanes **468** extending between the outer band **462** and the inner band **464**. The vanes **468** extend in a generally radial direction. The vanes **468**, and the turbine nozzle **460**, is a static component and the vanes **468** may be cured to direct (e.g., spin or swirl) the combustion gases to turn the turbines (e.g., drive the turbine blades) of the first stage of the HP turbine **116**. In this embodiment, the turbine section is a multi-stage turbine and these combustion gases will drive subsequent stages of the HP turbine **116** and the LP turbine **118**. The turbine nozzle **460** may, thus, also be referred to as a stage one nozzle (S1N). As discussed above the HP turbine **116** and the LP turbine **118**, among other things, drive the LP compressor **110** and HP compressor **112**.
(74) As noted above, we realized that when designing hydrogen fuel combustor to meet NOx emission targets, the combustor residence time needs to be reduced. We sized the combustor **401**, and more specifically, the combustor liner **420** for various gas turbine engines and flow rates. These different embodiments are shown below in Table 1 and were developed for different bypass ratios and thrust classes of engines, characterized by the core airflow. In particular, we considered the height H, also referred to as burner dome height, of the combustion chamber **430** and the length L, also referred to as burner length, of the combustion chamber. Diluents could be used to suppress the temperature, and, thus, NOx production, in the combustion chamber **430** when hydrogen is used as the fuel. With the combustor sized as described in these embodiments, hydrogen fuel can be used without the need of diluents. In some embodiments, no diluent is added to the combustion chamber

430 and the fuel is substantially completely diatomic hydrogen without diluent. As used herein, the term "substantially completely," as used to describe the amount of a particular element or molecule (e.g., diatomic hydrogen), refers to at least 99% by mass of the described portion of the element or molecule, such as at least 97.5%, such as at least 95%, such as at least 92.5%, such as at least 90%, such as at least 85%, or such as at least 75% by mass of the described portion of the element or molecule.

(75) FIGS. **6** to **10** illustrate how the height H and length L may be determined for the different shapes of combustion liners **420** shown in these figures. The height H of the combustion chamber **430** is taken at the forward end **432** of the combustion chamber **430**. The height H is the maximum height between an inner surface of the outer liner **422** and an inner surface of the inner liner **424** at the forward end **432** of the combustion chamber **430**. The height H is measured along a line (referred to as a forward line **472**, herein) that is generally orthogonal the inner surfaces of the outer liner **422** and the inner liner **424**. The forward line **472** may be orthogonal to a central axis **477** of the fuel nozzle assembly **440** and/or the fuel nozzle **442**. In this manner, the height H may be orthogonal to the central axis **477**. In some embodiments, the height H measured using with the forward line is the maximum height of the combustion chamber **430** and may also be the maximum dome height of the combustion chamber **430**.

(76) The length L of the combustion chamber **430** is the distance between forward line **472** and the leading edge **466** of the turbine nozzle **460**. As with the height H, a line (referred to as the aft line **474**, herein) can be drawn from the leading edge **466** at the outer liner **422** and leading edge at the inner liner **424**. Each of the forward line **472** and the aft line **474** has a midpoint (midpoint **476** and midpoint **478**, respectively) that is halfway between the outer liner **422** and the inner liner **424**. The length L can be measured from the midpoint **476** of the forward line **472** to the midpoint **478** of the aft line **474**. The midpoint **478** may be the midspan height of the turbine nozzle **460**.

(77) When developing a gas turbine engine, the interplay between components can make it particularly difficult to select or to develop one component during engine design and prototype testing, especially, when some components are at different stages of completion. For example, one or more components may be nearly complete, yet one or more other components may be in an initial or preliminary phase such that only one (or a few) design parameters are known. It is desired to arrive at what is possible at an early stage of design, so that the down selection of candidate optimal designs, given the tradeoffs, become more possible. Heretofore, the process has sometimes been more ad hoc, selecting one design or another without knowing the impact when a concept is first taken into consideration. For example, various aspects of the fan **126** design, the HP compressor **112** design, and/or the LP compressor **110** design may not be known, but such components impact the core air flow through the core air flow path **121**, and, thus, may influence the design of the combustion chamber **430**.

(78) We desire to narrow the range of configurations or combination of features that can yield favorable results given the constraints of the design, feasibility, manufacturing, certification requirements, etc., early in the design selection process to avoid wasted time and effort. During the course of the evaluation of different embodiments as set forth above, we, the inventors, discovered, unexpectedly, that there exists a relationship between the burner length and the burner dome height, which uniquely identifies a finite and readily ascertainable (in view of this disclosure) number of embodiments suitable for a particular architecture that can meet NOx emissions for hydrogen fuel and provide desired flame residence times. This relationship is referred to by the inventors as the combustor size rating (CSR) (in), and is defined according to the following relationship (1) between burner length L (in) and burner dome height H (in):

Combustor Size Rating (CSR)=(*L*).sub.2/(*H*) (1)

(79) As discussed further below, we have identified a range of the Combustor Size Ratings that enable a combustion chamber **430** to be designed for a gas turbine engine **20** using hydrogen fuel. This relationship is applicable over a wide range of thrust class and engine designs. Using this

unique relationship, a combustor **150** design can be developed early in the design process that meets NOx emissions targets and reduces engine weight for gas turbine engines using hydrogen fuel.

(80) Table 1 describes exemplary embodiments 1 to 24 identifying the CSR for various hydrogen fuel burning engines. The embodiments 1 to 24 may be engines with either rich burn combustors or lean burn combustors. Each of embodiments 1 to 24 burns hydrogen fuel. Embodiments 1 to 24 may represent any of the engines described with respect to FIGS. **1** to **5** and can be applied to any of the combustion chamber **430** shapes shown in FIGS. **6** to **10**. In Table 1, the CSR is determined based on the relationship (1) described above. A core air flow parameter (CAFP) (kN) is defined according to the following relationship (2) between thrust (kN) and bypass ratio, both at take off.

(81) $\text{CoreAirFlowParameter} = \frac{\text{Thrust}}{\text{BypassRatio}}$ (2)

The burner length is the length L identified with respect to FIGS. **6** to **10**, and in the embodiments 1 to 24 is between two inches and six inches. In embodiments 1 to 24, the burner length squared may be between six square inches and thirty-five square inches. The burner dome height is the height H identified with respect to FIGS. **6** to **10**, and in the embodiments 1 to 24 is between two and one half inches and six inches.

(82) TABLE-US-00001 TABLE 1 Combustor Core Air Flow Size Rating Parameter Thrust Bypass Embodiment (in) (kN) (kN) Ratio 1 4.30 38.16 332.39 8.71 2 6.67 49.85 254.26 5.10 3 6.67 53.44 272.53 5.10 4 6.67 51.18 261.03 5.10 5 6.67 52.36 267.03 5.10 6 4.69 21.07 120.10 5.70 7 4.69 23.80 121.40 5.10 8 3.01 12.58 64.53 5.13 9 3.01 12.18 62.49 5.13 10 3.00 16.44 83.70 5.09 11 3.00 15.20 82.10 5.40 12 3.00 14.27 84.20 5.90 13 3.00 14.27 84.20 5.90 14 3.00 14.27 84.20 5.90 15 2.12 36.55 321.60 8.80 16 2.12 39.23 345.20 8.80 17 2.12 40.65 349.20 8.59 18 2.12 40.65 349.20 8.59 19 2.12 37.34 299.81 8.03 20 1.67 13.63 143.10 10.50 21 1.94 51.51 489.30 9.50 22 5.51 40.89 363.90 8.90 23 2.46 12.72 147.28 11.58 24 2.70 5.00 150.00 30.00

(83) The length L may be between 2.63 inches and 5.60 inches. The length L may be between two inches and three inches. The length L may be between two and one half inches and three and one half inches. The height H may be between 2.80 inches and 5.60 inches. The height H may be between two and one half inches and six inches. The height H may be between two and one half inches and five inches. The height H may be between four inches and five inches. The burner length squared may be between 6.89 inches and 31.36 inches. The burner length squared may be between six square inches and thirty-five square inches. The burner length squared may be between six square inches and twenty square inches. The burner length squared may be between six square inches and twelve square inches. The burner length squared may be between eight square inches and twelve square inches. The burner length squared and the height may be any values such that the CSR is less than seven inches. The burner length squared and the height may be any values such that the CSR is less than six inches.

(84) FIG. **11** represents, in graph form, the burner length, squared, as a function of the burner dome height. FIG. **11** shows that the burner length, squared, may be changed based on the burner dome height. An area **500** may present the boundaries of burner length, squared, as a function of burner dome height in which a particular combustor is designed. FIG. **12** represents, in graph form, the CSR as a function of core air flow parameter. Table 1 and FIG. **12** show that CSR may be changed based on a thrust class, as characterized by the core air flow parameter, of an engine. An area **600** may present the boundaries of CSR as a function of the core air flow parameter in which a particular combustor is designed.

(85) As shown in FIG. **12**, the CSR is less than seven inches for every core air flow. That is, the CSR is less than seven inches for every thrust class of engine. The CSR may be between 1.67 inches and 6.67 inches. The CSR may be between one inch and seven inches. The CSR may be between one and one half inches and seven inches. The CSR may be between two inches and seven inches. The CSR may be between two inches and six inches. The CSR may be between one inches

and five inches. The CSR may be between two inches and five inches. The CSR may be between three inches and five inches. The core air flow parameter may be less than sixty kN. The core air flow parameter may be between five kN and 53.44 kN. The core air flow parameter may be between two and one half kN and sixty kN. The core air flow parameter may be between ten kN and twenty kN. The core air flow parameter may be between thirty kN and forty-five kN.
(86) With continued reference to FIG. **12**, the CSR may be a function of the core air flow parameter. The CSR may be based on a thrust of the gas turbine engine. The CSR may be between one inch and seven inches at a core air flow parameter between two and one half kN and sixty kN. The CSR may be between two inches and three and one quarter inches at a core air flow parameter between two and one half kN and fifty kN. The thrust may be between sixty kN and five hundred kN. The thrust may be between 62.49 kN and 489.30 kN. The CSR is defined by a relationship of the burner length, squared, and the burner dome height.
(87) In an extension of the concepts disclosed hereinabove, also provided herein are combustors that include a fuel-air mixing assembly fluidly coupled to the compressor section. The fuel-air mixing assembly includes an outer wall, a center body at least partially circumscribed by the outer wall, and an annular flow passage between the outer wall and center body. At least one fuel orifice includes a fuel outlet fluidly coupled to the annular flow passage.
(88) It has been determined that hydrogen fuel results in higher combustion temperatures as compared to combustible hydrocarbon liquid fuel and that NOx emissions are sensitive to combustor residence times. The present disclosure allows for fuel-air mixing assemblies that allow for efficient combustor usage and NOx emissions to be met. The fuel-air mixing assembly maintains velocity of the mixture of fuel and compressed air. More specifically, a constant passage area in the air passage downstream of the plurality of fuel orifices maintains well defined high velocity flow aft of injection to reduce or eliminate flame holding or flashback when using hydrogen fuels. The angle of the plurality of fuel orifices can be partially tangential to one or more portions of the airflow to improve mixing in the shortened mixing section. The plurality of fuel orifices can be inclined in the direction of the airflow to allow the fuel to follow the air velocity and reduce wakes due to fuel injection itself. The reduction of the wakes reduces flash back.
(89) Further, the fuel-mixing assembly as described herein, provides a structure that is designed for the heightened temperatures. When compared to traditional fuel-air mixing assemblies, the fuel-mixing assembly includes fuel orifices having customizable characteristics, which can improve the uniformity of the fuel-air mixture. The various embodiments of the combustors, as described herein and shown in the figures, include a plurality of fuel orifices capable of injecting fuel in a low turbulence region, which can provide for a shorter mixing length resulting in reduced flame holding. Fuel can be injected from sets of fuel orifices on both the center body and the outer wall to achieve better mixing and control the fuel, improving fuel penetration circumferentially into the airflow. Once the fuel-air mixture passes the axial or second end of the center body and is ignited, the centrally located fuel-air mixture provides a lifted flame, which also reduces the chance of flame holding. The described fuel-mixing assembly is particularly beneficial when utilizing hydrogen fuel, which burns hotter than traditional fuels, as the assembly permits an increase in fuel efficiency and allows for NOx emissions to be met.
(90) Referring now to FIG. **13** a schematic gas turbine engine **710** is illustrated as having a compressor section **712**, a combustion section **714**, and a turbine section **716** in an axial flow arrangement. A drive shaft **718** rotationally couples the compressor section **712** and turbine section **716**, such that rotation of one affects the rotation of the other, and defines a rotational axis **720** for the gas turbine engine **710**. The combustion section **714** can be fluidly coupled to at least a portion of the compressor section **712** and the turbine section **716** such that the combustion section **714** at least partially fluidly couples the compressor section **712** to the turbine section **716**. The gas turbine engine **710** can be similar to the turbofan engine **100** as previously described.
(91) A low-pressure compressor **722** and a high-pressure compressor **724** serially fluidly couple to

one another. The combustion section **714** can be fluidly coupled to the high-pressure compressor **724** at an upstream end of the combustion section **714** and to a high-pressure turbine **726** at a downstream end of the combustion section **714**.

(92) During operation, ambient or atmospheric air is drawn into the compressor section **712**, where the air is compressed defining a pressurized air. The pressurized air can then flow into the combustion section **714** where the pressurized air is mixed with fuel in a fuel-air mixing assembly **800** located in a combustor **730**, and ignited, thereby generating combustion gases. Some work is extracted from these combustion gases by the high-pressure turbine **726**, which can drive at least the high-pressure compressor **724**. The combustion gases are discharged into the low-pressure turbine **728**, which can extract additional work to drive the low-pressure compressor **722**, and the exhaust gas is ultimately discharged from the gas turbine engine **710**.

(93) FIG. **14** illustrates a cross-sectional view of a fuel-air mixing assembly **800**, which can be utilized within the combustion section **714**. The fuel-air mixing assembly **800** can be part of a fuel nozzle located upstream and fluidly coupled to the combustion section **714** or the combustor **730**. It will be understood that the fuel-air mixing assembly **800** can be utilized with any of the described combustors and engines.

(94) The fuel-air mixing assembly **800** includes at least an outer wall **802**, a center body **804**, an annular flow passage or air passage **806**, a plurality of apertures **808**, a fuel cavity **810**, and a plurality of fuel orifices **812**. It will be understood that the outer wall **802**, the center body **804**, the air passage **806**, and the fuel cavity **810**, can be circumferential, circular, or annular about a center body axis or centerline **842**.

(95) The outer wall **802** can be a combustor liner, shroud, or a mixing tube outer wall, in non-limiting examples. The outer wall **802** includes an outer wall inner surface **820** and an outer wall outer surface **822**. An outer wall thickness can be defined between the outer wall inner surface **820** to the outer wall outer surface **822**. A plurality of apertures **808** extend through the outer wall **802** and pass from the outer wall inner surface **820** to the outer wall outer surface **822**.

(96) The center body **804** can be at least partially circumscribed by the outer wall **802**. The center body **804** can have a center body outer surface **824** and a center body inner surface **826**. The center body **804** is illustrated as being at least partially hollow. The center body **804** can extend from the outer wall **802** at a fore end or first end **828** to an aft end or second end **830**. The second end **830** of the center body **804** can be the axially farthest point or end of the center body **804** extending downstream. The second end **830** can be at an exit plane, at which point the fuel-air mixture would exit the portion of the air passage **806** partially defined by the center body **804**. Alternatively, the second end **830** can be the end of the center body **804** that is axially downstream of the first end **828**, wherein the second end **830** is circumscribed by the outer wall **802**. A center body length **831** can be measured axially from the first end **828** to the second end **830**.

(97) A recess distance **832** can be measured from the second end **830** to a peak recess point **834**. The recess distance **832** can be between or equal to 0.0 centimeters to 1.3 centimeters. Optionally, the recess distance **832** can be between or equal to 0.3 centimeters to 0.9 centimeters.

(98) Additionally, or alternatively, the recess distance **832** can be between or including 0%-25% of the center body length **831** or 0% to 100% of a diameter of the center body **804** at the second end **830**. While illustrated as concave, the second end **830** can have a shape that is concave, convex, planer, or any combination therein. Further still, while the term diameter is utilized in discussion herein it will be understood that the profile, shape, etc. of the described elements, apetures, and passages need not be circular. Any suitable shape will suffice.

(99) The air passage **806** can be at least partially defined by the outer wall **802** and the center body **804**. An inlet **807** to the air passage **806** can be located at or adjacent the first end **828** of the center body **804**. An outlet **809** of the air passage **806** can be located at or adjacent the second end **830** of the center body **804**. More specifically, the air passage **806** can be defined as the space between the outer wall inner surface **820** and the center body outer surface **824**. An air passage area can be

defined as the area of a cross section of the air passage **806**. The air passage area can be proportional to an air passage diameter **836**, wherein the air passage diameter **836** is a distance measure from the outer wall inner surface **820** to the center body outer surface **824**. Optionally, the air passage area of the air passage **806** increases as the air passage diameter **836** increases, decreases as the air passage diameter **836** decreases, and remains constant when the air passage diameter **836** remains constant.

(100) The plurality of apertures **808** can extend through the outer wall **802** and fluidly couple the compressor section **712** (FIG. **13**) to the air passage **806**. A first set of apertures **808***a* and a second set of apertures **808***b* can be defined by the plurality of apertures **808**. The first set of apertures **808***a* can be circumferentially spaced and generally located at a first axial position. The second set of apertures **808***b* can be circumferentially spaced apertures and generally located at a second axial position, wherein the second axial position in a non-limiting example is downstream of the first axial position relative to an airflow through the air passage **806**. That is, the centerline of each aperture of the first set of apertures **808***a* is upstream of a centerline for each aperture of the second set of apertures **808***b*.

(101) The fuel cavity **810** is defined, at least in part, by the center body **804**. More specifically, the center body inner surface **826** can define a hollow portion, which is the fuel cavity **810**. It is contemplated that the fuel cavity **810** can be a hydrogen fuel cavity where the hydrogen fuel cavity can provide hydrogen-containing fuel to one or a plurality of fuel orifices **812**. The fuel cavity **810** can include at least one channel such as the channel **840**, which is defined within the center body **804**, radially exterior of the center body inner surface **826**. The channel **840** is fluidly coupled to the fuel cavity **810** and can extend in an aft-to-fore direction. A channel outer surface **846** and a channel inner surface **848** can define the channel **840**. The channel **840** can receive fuel from the fuel cavity **810** at an inlet **844**. The channel **840** can have one or more portions that extend towards the center body outer surface **824**. Additionally, or alternatively, the channel **840** can include one or more portions that extend parallel to the center body outer surface **824** or the center body inner surface **826**. It is contemplated that the channel **840** can have one or more changes in direction relative to the centerline **842** of the fuel cavity **810**. It is contemplated that the diameter of the channel **840** can remain constant or include one or more portions in which the diameter is changing. The channel **840** can extend an axial distance between 0% to 75% of the center body length **831**, however it is contemplated that the channel **840** can extend an axial distance between 2% to 50% of the center body length **831**. A protrusion **850** can extend from a portion of the center body inner surface **826**. The protrusion **850** can have a uniform thickness although it need not and can instead have one or more portions in which the thickness changes, either continuously or discretely.

(102) The plurality of fuel orifices **812** fluidly couple the channel **840** to the air passage **806**. The plurality of fuel orifices **812** can be circumferentially located about the center body **804**. A fuel inlet **852** can be located at the channel outer surface **846** such that the fuel inlet **852** can receive fuel from the channel **840**. A fuel outlet **854** is located at the center body outer surface **824** to provide fuel to the air passage **806**. While the injection diameter of the plurality of fuel orifices **812** has been illustrated as constant, it will be understood that this need not be the case. The injection diameter change in one or more portions as the plurality of fuel orifices **812** extend radially outward from the fuel cavity **810** to the air passage **806**. It is further contemplated that the injection diameter need not be the same between a first orifice of the plurality of fuel orifices **812** and a second orifice of the plurality of fuel orifices **812**.

(103) The plurality of fuel orifices **812** can be located at a third axial position. More specifically, a fuel orifice centerline **856** of the fuel outlet **854** can be located at least 0.5 centimeters from the second set of apertures **808***b* or an aperture centerline **857** of the second set of apertures **808***b*, defining an orifice distance **858**. Additionally, or alternatively, the orifice distance **858** can be between or equal to 10%-95% of the center body length **831**.

(104) A predetermined distance or fuel orifice distance **860** can be measured from the fuel orifice centerline **856** to the second end **830**. The fuel orifice distance **860** can be between or equal to 0.0 centimeters and 2.0 centimeters. It is contemplated that the fuel orifice distance **860** can be between or equal to 0%-50% of the center body length **831**. Additionally, or alternatively, the fuel orifice distance **860** can be between or equal to 0% to 100% of the diameter of the center body **804** at the second end **830**.
(105) A constant cross-sectional area portion or constant area portion **862** of the air passage **806** can be located between the plurality of fuel orifices **812** and the second end **830**. That is, the air passage diameter **836** is constant between at least a first point **864** downstream of the plurality of fuel orifices **812** and a second point **866** downstream of the first point **864**, wherein the second point **866** is at the second end **830**, as illustrated, or upstream of the second end **830**. Stated another way, the constant area portion **862** has a constant cross-sectional area along a predetermined portion of the center body **804**, starting at the first point **864** and terminating at the second end **830**. It is contemplated that the fuel outlet **854** is located at or opens into the constant area portion **862**.
(106) FIGS. **15**A-**15**E illustrate, non-limiting profiles or shapes that at least one fuel orifice of the plurality of fuel orifices **812** can take. A circular profile **868** of at least one fuel orifice of the plurality of fuel orifices **812** is shown in FIG. **15**A. It is contemplated that such a circular profile can include a perfect circle or include radius measurements within 5% of each other. A triangular profile **870** of at least one fuel orifice of the plurality of fuel orifices **812** is shown in FIG. **15**B. The triangular shape can be any triangle, including, but not limited to an equilateral triangle, an acute triangle, a right triangle, or an obtuse triangle. Additional or alternatively, one or more legs or angles of the triangle can be equal or have measurements within 5% of each other. A squoval profile **872** of at least one fuel orifice of the plurality of fuel orifices **812** is illustrated in FIG. **15**C. This can also be considered a stadium shape, race track shape, a rounded rectangle, or any rectangle with chamfered corners. A teardrop profile **874** of at least one fuel orifice of the plurality of fuel orifices **812** is illustrated in FIG. **15**D. The teardrop shape, or lachrymiform, can have a rounded smaller portion and a rounded larger portion, as illustrated, or include a smaller pointed portion and a larger rounded portion. An oval profile **876** of at least one fuel orifice of the plurality of fuel orifices **812** is illustrated in FIG. **15**E. The oval profile may include any elliptical shape including an ellipse, sub elliptical, pyriform, or any combination therein. It will be understood that the profiles described herein may be utilized separately or in combination within the fuel-air mixing assembly **800**.
(107) With reference to FIGS. **13-14**, in operation, an airflow from the high-pressure compressor **724** flows through the plurality of apertures **808** into the air passage **806**. A steady airflow is developed in the air passage **806**. Once the steady airflow is established, fuel, for example hydrogen-containing fuel, from the fuel cavity **810** flows into the channel **840** via the inlet **844**. Fuel in the channel **840** then flows into the plurality of fuel orifices **812** via the fuel inlet **852**. At the fuel outlet **854** fuel is introduced or injected to the airflow in the air passage **806**. The fuel is introduced to the airflow in the air passage **806** in a low turbulent region, which helps to reduce flame holding. The plurality of fuel orifices **812** are spread circumferentially to provide uniform fuel spread, resulting in better mixing and at the same time achieving fuel penetration into the airflow such that the fuel-air mixture stays away from the outer wall **802** or the center body **804**. The plurality of fuel orifices **812** are located 2.0 centimeters or less from the aft end of the center body **804**. The location of the plurality of fuel orifices **812** helps to reduce flame holding at the center body **804** or the air passage **806**.
(108) The air passage **806** includes the constant area portion **862** that helps to maintain high velocity of the air-fuel mixture. That is, the constant area portion **862** can maintain a high velocity of the air-fuel mixture over a longer length than existing designs for fuel-air mixing. The high velocity of the air-fuel mixture reduces flash back into the air passage **806**, allowing the gas turbine engine **710** to utilize hydrogen-containing fuel, which burns hotter than traditional fuels.

(109) The air-fuel mixture is combusted downstream of the center body **804**. Due to uniform mixing of the fuel with the air, upon combustion, the temperature distribution in the combustion section **714** or the combustor **730** is more uniform, permitting the use of higher-temperature fuels, such as hydrogen, which provides for reducing or eliminating emissions, while maintaining or improving engine efficiency.
(110) FIG. **16** illustrates a fuel-air mixing assembly **900** by way of further non-limiting example. The fuel-air mixing assembly **900** is similar to the fuel-air mixing assembly **800** of FIG. **15**, and it will be understood that the description of the like parts applies unless otherwise noted. The fuel-air mixing assembly **900** includes at least the outer wall **802**, the center body **804**, the air passage **806**, the plurality of apertures **808**, the fuel cavity **810**, and a plurality of fuel orifices **912**.
(111) The plurality of fuel orifices **912** can include a first set of fuel orifices **912***a* and a second set of fuel orifices **912***b*. The first set of fuel orifices **912***a* can be axially spaced from the second set of fuel orifices **912***b*. That is, the first set of fuel orifices **912***a* and the second set of fuel orifices **912***b* can be staggered axially to achieve better fuel spread and intermixing with the air supply. It is further contemplated that each of the fuel orifices within the first set of fuel orifices **912***a* or the second set of fuel orifices **912***b* can also vary in axial location in relationship to fuel orifices from the same set. The first set of fuel orifices **912***a* and the second set of fuel orifices **912***b* can be an axial distance between 0.0 centimeters and 2.0 centimeters from the second end **830**. A first outlet **954***a* (see FIG. **17**) of the first set of fuel orifices **912***a* can be radially offset from at least one second outlet **954***b* (see FIG. **18**) of the second set of fuel orifices **912***b*. It is contemplated that the distance between the first set of fuel orifices **912***a* and the second set of fuel orifices **912***b* can be 0-30% of the diameter of the center body **804** at the second end **830**. The axial staggering of the first set of fuel orifices **912***a* and the second set of fuel orifices **912***b* can further improve the distribution of fuel to the airflow in the air passage **806**, and can improve mixing of the fuel and air supplies prior to a constant area portion **962**.
(112) In a non-limiting example, the outer wall inner surface **820** can include one or more bumps, projections, or protrusions, illustrated, by way of example, by a reduction portion **901**. The reduction portion **901** can be unitarily formed with the outer wall **802**. Alternatively, the reduction portion **901** can be material coupled to the outer wall inner surface **820**. The reduction portion **901** can include a reducing cross-sectional area portion or converging portion upstream of the plurality of fuel orifices **912**. Downstream of the plurality of fuel orifices **912**, the reduction portion **901** can maintain the constant area portion **962**, although at a smaller air passage diameter and air passage area than without the reduction portion **901**. The reduction portion **901** can increase airflow speed at the converging portion upstream of the fuel injection and maintain that speed through the constant area portion **962** downstream of the plurality of fuel orifices **912**. These higher, maintained velocities over a longer axial length for the fuel-air mixture can prevent flash back.
(113) It is contemplated that the reduction portion **901** can include a converging, sloped, or angled portion that extends axially to within a distance of the fuel outlet **854** that is 10% or less of the diameter of the center body **804** at the second end **830**. Downstream of the fuel outlet **854**, the reduction portion **901** can have a cylindrical or constant nominal diameter portion. In a further non-limiting example, the angled portion can extend axially to or beyond the fuel outlet **854** or second end **830**.
(114) It is contemplated that the center body **804**, as illustrated, can have an increasing diameter portion, such that the center body outer surface **824** narrows or reduces the diameter of the air passage **806**. It is further contemplated, in a non-limiting example, that the center body **804** can optionally include a cylindrical section **903**, or constant diameter portion. The constant diameter portion of the center body **804** can axially overlap the converging portion or the constant nominal diameter portion of the reduction portion **901**. It is also contemplated that the fuel outlet **854** can be located in the constant diameter portion of the center body **804**. That is, the constant diameter portion of the center body **804** can extend upstream, downstream, or a combination of upstream

and downstream of the fuel outlet **854**. Additionally, or alternatively the constant diameter portion of the center body **804** can extend between 10%-100% of the center body length. It is further contemplated, in a non-limiting example, that the increasing diameter portion of the center body **804** can extend to the second end **830**. That is, the increasing diameter portion of the center body **804** can be 5%-100% of the center body length.

(115) FIG. **17** is cross section taken along line XVII-XVII of FIG. **16** further illustrating the first set of fuel orifices **912***a*. The first set of fuel orifices **912***a* extend from the channel inner surface **848** to the center body outer surface **824**. That is, the first set of fuel orifices **912***a* fluidly couple the channel **840** with the air passage **806**. A first angle **908** can be defined as the angle between a fuel orifice centerline **956***a* and a vertical reference line **978***a*. The vertical reference line **978***a* is perpendicular to the centerline **842** of the fuel cavity **810**. As illustrated, the first angle **908** can be a non-zero angle, however any angle, including zero is contemplated. The first outlet **954***a* of the first set of fuel orifices **912***a* is circumferentially offset from the second outlet **954***b* of the second set of fuel orifices **912***b*. Such an offset can improve uniform fuel distribution and mixing of the fuel and air.

(116) FIG. **18** is cross section taken along line XVIII-XVIII of FIG. **16** further illustrating the second set of fuel orifices **912***b*. The second set of fuel orifices **912***b* extend from the channel inner surface **848** to the center body outer surface **824**. That is, the second set of fuel orifices **912***b* fluidly couple the channel **840** with the air passage **806**. A second angle can be defined as the angle between a fuel orifice centerline **956***b* and a vertical reference line **978***b*. The vertical reference line **978***b* is perpendicular to the centerline **842** of the fuel cavity **810** and in the same plane as the vertical reference line **978***b*. As illustrated, the second angle can be zero, as the fuel orifice centerline **956***b* is aligned with the vertical reference line **978***b*, such that they are shown as overlapping, however any non-zero angle is also contemplated. This also illustrates a situation where the first outlet **954***a* of the first set of fuel orifices **912***a* is circumferentially offset from the second outlet **954***b* of the second set of fuel orifices **912***b*, the offset can improve uniform fuel distribution and mixing of the fuel and air.

(117) FIG. **19** is a non-limiting example of a plurality of fuel orifices **1012**, which can be utilized. The plurality of fuel orifices **1012** are similar to the plurality of fuel orifices **912**, **912***a*, **912***b*, therefore, like parts will be identified with like numerals increased by 100, with it being understood that the description of the like parts of the plurality of fuel orifices **912**, **912***a*, **912***b* applies to the plurality of fuel orifices **1012**.

(118) The plurality of fuel orifices **1012** extend from the channel inner surface **848** to the center body outer surface **824**. That is, the plurality of fuel orifices **1012** fluidly couple the channel **840** with the air passage **806**. An orifice angle **1084** can be defined as the angle between a first radius **1086** extending from the centerline **842** through an inlet **1052** and a second radius **1088** extending from the centerline **842** through an outlet **1054**. As illustrated, the orifice angle can be non-zero. In one example, the orifice angle **1084** can be between or equal to −60 degrees to 60 degrees, that is 60 degrees counter clockwise to 60 degrees clockwise. It is further contemplated that the orifice angle **1084** can be between 0-30 degrees, although any angle value, including zero, is also contemplated.

(119) By way of non-limiting example, the plurality of fuel orifices **1012** can include at least one diverter **1090**. The at least one diverter **1090** can change the direction of the flow, limit the volume of the flow, increase or decrease the speed of the flow, or change the direction of the flow, or even increase or decrease local turbulence. Additionally, one or more valves (not show for clarity) can be included in one or more of the plurality of fuel orifices **1012**.

(120) An orifice centerline angle **1085** can be measured from the first radius **1086** extending from the centerline **842** through the inlet **1052** and a fuel orifice centerline **1056**. The orifice centerline angle **1085** can be, for example, between or equal to −60 degrees to 60 degrees. That is, the orifice centerline angle **1085** can be 60 degrees counter clockwise to 60 degrees clockwise. It is further

contemplated that the orifice centerline angle **1085** can be between 0-30 degrees, although any angle value, including zero, is also contemplated.
(121) FIG. **20** illustrates a fuel-air mixing assembly **1100**. The fuel-air mixing assembly **1100** is similar to the fuel-air mixing assembly **800**, **900** therefore, like parts will be identified with like numerals increased by 200, with it being understood that the description of the like parts of the fuel-air mixing assembly **800**, **900** applies to the fuel-air mixing assembly **1100**, unless otherwise noted. The fuel-air mixing assembly **1100** includes at least the outer wall **802**, the center body **804**, the air passage **806**, the plurality of apertures **808**, the fuel cavity **810**, and a plurality of fuel orifices **1112**.
(122) The outer wall **802** includes the outer wall inner surface **820** and the outer wall outer surface **822**. The plurality of apertures **808** extend through the outer wall **802** from the outer wall inner surface **820** to the outer wall outer surface **822**.
(123) The center body **804** can be at least partially circumscribed by the outer wall **802**, having the second end **830** of the center body **804** as the farthest point or end of the center body **804** extending downstream within the outer wall **802**. The fuel cavity **810** is defined, at least in part, by the center body **804**.
(124) At least one channel or channel **1140** can extend from the fuel cavity **810** into the outer wall **802** upstream of the center body **804**. The channel **1140** can curve, bend, or otherwise include any shape that allows the channel **1140** to be defined within the outer wall **802**. That is, the channel **1140** and the plurality of apertures **808** do not intersect. The channel **1140** can have one or more portions that extend towards the outer wall outer surface **822**. Additionally, or alternatively, the channel **1140** can include one or more portions that extend parallel to the outer wall inner surface **820**. The channel **1140** can fluidly couple the fuel cavity **810** to another fuel cavity illustrated as at least one fuel tank **1194**. The fuel tank **1194** is illustrated as defined by the outer wall **802**, however, it is contemplated that the fuel tank **1194** can be coupled to the outer wall **802**.
(125) The plurality of fuel orifices **1112** fluidly couple the fuel cavity **810** to the air passage **806**. As illustrated, by way of example, the plurality of fuel orifices **1112** fluidly couple the fuel tank **1194** to the air passage **806**. The plurality of fuel orifices **1112** can be circumferentially spaced about the center body **804**. A fuel inlet **1152** receives fuel into at least one fuel orifice of the plurality of fuel orifices **1112** from the fuel cavity **810** via the channel **1140** and the fuel tank **1194**.
(126) A fuel outlet **1154** can be located at the outer wall inner surface **820** to provide fuel to the air passage **806**. It is contemplated that the injection diameter of the plurality of fuel orifices **1112** can be constant, as illustrated, or change in one or more portions of the fuel orifice as the plurality of fuel orifices **1112** extend radially outward. It is further contemplated that the injection diameter can vary between two or more fuel orifices of the plurality of fuel orifices **1112**.
(127) It is contemplated that the channel **1140** or the fuel tank **1194** can be a hydrogen channel or hydrogen fuel tank where the hydrogen channel or the hydrogen fuel tank can provide hydrogen-containing fuel to at least one fuel orifice. In addition to or in place of the channel **1140**, an outside fuel source **1113** can be coupled to the fuel tank **1194**. The outside fuel source **1113** can include any number or combination of additional tanks, pump, conduits, or valves. It is contemplated that the outside fuel source **1113** can be a hydrogen outside fuel source where the hydrogen outside fuel source can provide hydrogen-containing fuel to at least one fuel orifice.
(128) The plurality of fuel orifices **1112** can be located downstream of the plurality of apertures **808**. That is, a fuel orifice centerline **1156** of the fuel outlet **1154** can be located at least 0.5 centimeters from the plurality of apertures **808**. In other words, an aperture to orifice distance **1158** can be equal to or more than 0.5 centimeters. Additionally, or alternatively, the aperture to orifice distance **1158** can be between or equal to 10%-95% of a center body length. A fuel orifice distance **1160** can be measured from the fuel orifice centerline **1156** to the second end **830**. The fuel orifice distance **1160** can be between or equal to 0.0 centimeters and 2.0 centimeters. It is contemplated that the fuel orifice distance **1160** can be between or equal to 0%-50% of the center body length. It

is also contemplated that the fuel orifice distance **1160** can be between or equal to 0%-100% of the diameter of the center body **804** measured at the second end **830**.

(129) A constant area portion **1162** of the air passage **806** can be located between the plurality of fuel orifices **1112** and the second end **830**. That is, the air passage diameter **1136** is constant between at least a first point **1164** downstream of the plurality of fuel orifices **812** and a second point **1166** downstream of the first point **1164**, wherein the second point **1166** is at the second end **830**, as illustrated, or upstream of the second end **830**. The constant area portion **1162** imparts a high velocity component to the mixture of air and gas emitted from the fuel orifice assembly, while the channel **1140** provides for injecting fuel radially inward, as opposed to radially outward as shown FIGS. **14** and **16**. A radially inward injection can provide for improved fuel and air mixing prior to the constant area portion **1162**.

(130) FIG. **21** further illustrates a portion of another fuel-air mixing assembly **1200**. The fuel-air mixing assembly **1200** is similar to the fuel-air mixing assembly **700**, **800**, **1100** therefore, like parts will be identified with like numerals increased by 100 with it being understood that the description of the like parts of the fuel-air mixing assembly **700**, **800**, **1100** applies to the fuel-air mixing assembly **1200**, unless otherwise noted. The fuel-air mixing assembly **1200** includes at least the outer wall **802** with the plurality of apertures (not shown), the center body **804**, the air passage **806**, the fuel cavity **810**, and a plurality of fuel orifices **1212**.

(131) The plurality of fuel orifices **1212** include a first set of fuel orifices **1212***a* and a second set of fuel orifices **1212***b*. The first set of fuel orifices **1212***a* pass through at least a portion of the outer wall **802**. The first set of fuel orifices **1212***a* fluidly couple the air passage **806** with a fuel tank (not shown) or other fuel source to provide fuel to the air passage **806**. An inlet **1252***a* can be fluidly coupled to an outlet **1254***a* via the first set of fuel orifices **1212***a*.

(132) The second set of fuel orifices **1212***b* pass through a portion of the center body **804**. That is, the second set of fuel orifices **1212***b* can be radially spaced from the first set of fuel orifices **1212***a*. The second set of fuel orifices **1212***b* fluidly couple the channel **840** with the air passage **806** to provide fuel to the air passage **806**, wherein the channel **840** is fluidly coupled to the fuel cavity **810**. An inlet **1252***b* can be fluidly coupled to an outlet **1254***b* via the second set of fuel orifices **1212***b*. The plurality of fuel orifices **1212** can be located in different axial positions. That is, the first set of fuel orifices **1212***a* can be at a different axial location that the second set of fuel orifices **1212***b*. Additionally or alternatively, the orifices within the first or second set of fuel orifices **1212***a*, **1212***b* can be located at a variety of axial location, wherein the axial location is not uniform through each set.

(133) Optionally, protrusion passages **1255** can fluidly couple the channel **840** with the fuel cavity **810**. The protrusion passages **1255** can have similar characteristics to the plurality of fuel orifices **1212**. That is, the protrusion passages **1255** can be circumferentially spaced, angled axially, or angled circumferentially. Further, the protrusion passages **1255** can have any shape, including a changing shape cross section.

(134) While illustrated as axially aligned with the second set of fuel orifices **1212***b*, it is contemplated that the protrusion passages **1255** can be at any axial location in alignment with, upstream, or downstream of the first set of fuel orifices **1212***a* or the second set of fuel orifices **1212***b*. Additionally or alternatively, the protrusion passages **1255** can be located at a variety of axial locations with respect to other protrusion passages **1255**. That is, the axial location does not have to be uniform for all protrusion passages **1255**. Optionally, the protrusion **850** can extend to a downstream end portion **1257** of the fuel cavity **810**. In this example, the protrusion passages **1255** would fluidly couple the fuel cavity **810** and the channel **840**.

(135) A fuel orifice distance **1260** can be measured from a fuel orifice centerline **1256** to the second end **830** of the center body **804**. The fuel orifice distance **1260** can be between or equal to 0.0 centimeters and 2.0 centimeters. It is contemplated that the fuel orifice distance **1260** can be between or equal to 0%-50% of the center body length. It is further contemplated that the fuel

orifice distance **1260** can be between or equal to 0%-100% of the diameter of the center body **804** at the second end **830**. In a non-limiting example, even if the first set of fuel orifices **1212***a* and the second set of fuel orifices **1212***b* do not axial align, that the distance between each orifice of the first set of fuel orifices **1212***a* and the second set of fuel orifices **1212***b* have a fuel orifice distance equal to or less than 2.0 centimeters or 0%-50% of the center body length.

(136) A constant area portion **1262** of the air passage **806** can be located between the plurality of fuel orifices **1212** and the second end **830**. That is, an air passage diameter **1236** is constant between at least a first point **1264** downstream of the plurality of fuel orifices **1212** and a second point **1266** downstream of the first point **1264**, wherein the second point **1266** is at the second end **830**, as illustrated, or upstream of the second end **830**.

(137) In operation, one or both of the first set of fuel orifices **1212***a* or the second set of fuel orifices **1212***b* can be used to provide fuel to the air passage **806**. The contribution or activation of one or more or one or more sets of the plurality of fuel orifices **1212** allows for fuel injection from both the center body **804** and the outer surface or the outer wall **802**. Providing fuel from more than one radial location can improve control of the mixing of the fuel from the plurality of fuel orifices **1212** and the airflow in the air passage **806**. This can improve engine response, as different fuel-air mixtures are needed during different portions of a cycle of operation of the turbine engine **710**.

(138) Similarly, when fuel is from the plurality of fuel orifices **1212** on the center body **804** and the outer wall **802** is provided to an airflow in the air passage **806**, there is better fuel penetration circumferentially, as the fuel is added radially from the outside and inside of the airflow. This helps to keep the fuel-air mixture in the center of the air passage **806**. When the fuel-air mixture is centered in the air passage **806**, when the fuel-air mixture is ignited downstream of the center body **804**, a lifted flame is provided. That is, the flame is spaced from the center body **804**. Having a lifted flame further prevents flame holding and flashback.

(139) FIG. **22** is cross section along line XXII-XXII of FIG. **21** at the fuel orifice centerline **1256** further illustrating the first set of fuel orifices **1212***a* and the second set of fuel orifices **1212***b*. The first set of fuel orifices **1212***a* extend through a portion of the outer wall **802** to the outer wall inner surface **820**. That is, the first set of fuel orifices **1212***a* fluidly couple the fuel source with the air passage **806**. The second set of fuel orifices **1212***b* extend through a portion of the center body outer surface **824** to the center body inner surface **826**. That is, the second set of fuel orifices **1212***b* fluidly couple the air passage **806** with the fuel cavity **810**. An orifice set angle **1209** can be defined as the angle between a centerline of at least one orifice of the first set of fuel orifices **1212***a* and a centerline of at least one orifice of the second set of fuel orifices **1212***b*, where the centerlines are drawn extending from the centerline **842** of the fuel cavity **810**. As illustrated, the orifice set angle **1209** can be a non-zero angle, however any angle, including zero is contemplated, which is illustrated in FIG. **23**. It is contemplated that the angle need not equal between adjacent pairs of fuel orifices and that the orifices of the first set of fuel orifices **1212***a* and the second set of fuel orifices **1212***b* need not to be uniformly distributed about the circumference of the center body **804** or the outer wall **802**.

(140) Optionally, the protrusion passages **1255** can fluidly couple the channel **840** with the fuel cavity **810**. The protrusion passages **1255** can align with one or more of the first set of fuel orifices **1212***a* or the second set of fuel orifices **1212***b*. Alternatively, the protrusion passages **1255** can be form a non-zero angle with both the first set of fuel orifices **1212***a* and the second set of fuel orifices **1212***b*. That is, there can be any number of protrusion passages **1255** that can be circumferentially located at any location in alignment with or between the first set of fuel orifices **1212***a* or the second set of fuel orifices **1212***b*.

(141) FIG. **24** is another variation of the cross section of FIG. **22** taken at the fuel orifice centerline **1256** (FIG. **21**) further illustrating the first set of fuel orifices **1212***a* and the second set of fuel orifices **1212***b*.

(142) A first orifice angle **1284***a* for the first set of fuel orifices **1212***a* can be defined as the angle

between a first radius **1286***a* extending from the centerline **842** through the outlet **1254***a* and a second radius **1288***a* extending from the centerline **842** through the inlet **1252***a*. As illustrated, the orifice angle can be non-zero. Any angle value, including zero, is also contemplated.

(143) A second orifice angle **1284***b* for the second set of fuel orifices **1212***b* can be defined as the angle between a first radius **1286***b* extending from the centerline **842** through the inlet **1252***b* and a second radius **1288***b* extending from the centerline **842** through the outlet **1254***b*. As illustrated, the orifice angle can be non-zero. Any angle value, including zero, is also contemplated.

(144) As illustrated, by way of example, the clockwise or counter clockwise angle of the first set of fuel orifices **1212***a* can be opposite that of the second set of fuel orifices **1212***b*. It is contemplated that first orifice angle **1284***a* or the second orifice angle **1284***b* can be between or equal to −60 degrees to 60 degrees. That is, 60 degrees counter clockwise to 60 degrees clockwise. It is further contemplated that the first set of fuel orifices **1212***a* can be between or equal to zero degrees and 30 degrees.

(145) A first centerline angle **1285***a* for the first set of fuel orifices **1212***a* can be defined as the angle between the first radius **1286***a* extending from the centerline **842** through the outlet **1254***a* and a first fuel orifice centerline **1256***a*. As illustrated, the first centerline angle **1285***a* can be non-zero. Any angle value, including zero, is also contemplated.

(146) A second centerline angle **1285***b* for the second set of fuel orifices **1212***b* can be defined as the angle between the first radius **1286***b* extending from the centerline **842** through the inlet **1252***b* and a second fuel orifice centerline **1256***b*. As illustrated, the second centerline angle **1285***b* can be non-zero. Any angle value, including zero, is also contemplated.

(147) As illustrated, by way of example, the clockwise or counter clockwise centerline angle of the first set of fuel orifices **1212***a* can be opposite that of the second set of fuel orifices **1212***b*. It is contemplated that first centerline angle **1285***a* or the second centerline angle **1285***b* can be between or equal to −60 degrees to 60 degrees. That is, 60 degrees counter clockwise to 60 degrees clockwise.

(148) FIG. **25** illustrates another cross section of a portion of a fuel-air mixing assembly **1300**. The fuel-air mixing assembly **1300** is similar to the fuel-air mixing assembly **700**, **800**, **1100**, **1200** therefore, like parts will be identified with like numerals increased by 100 with it being understood that the description of the like parts of the fuel-air mixing assembly **700**, **800**, **1100**, **1200** applies to the fuel-air mixing assembly **1300**, unless otherwise noted. The fuel-air mixing assembly **1300** includes at least the outer wall **802** with the plurality of apertures (not shown), the center body **804**, the air passage **806**, the fuel cavity **810**, and a plurality of fuel orifices **1312**.

(149) The plurality of fuel orifices **1312** include a first set of fuel orifices **1312***a* and a second set of fuel orifices **1312***b*. The first set of fuel orifices **1312***a* pass through at least a portion of the outer wall **802**. The first set of fuel orifices **1312***a* fluidly couple the air passage **806** with a fuel tank (not shown) or other fuel source to provide fuel to the air passage **806**. The second set of fuel orifices **1312***b* pass through a portion of the center body **804**. The second set of fuel orifices **1312***b* fluidly couple the channel **840** with the air passage **806** to provide fuel to the air passage **806**, wherein the channel **840** is fluidly coupled to the fuel cavity **810**.

(150) A first angle **1311** can be defined as the angle between a reference line **1357** and a fuel orifice centerline **1359** of at least one of the fuel orifices of the first set of fuel orifices **1312***a*. The reference line **1357** can be parallel to the centerline **842** of the fuel cavity **810** or the centerline of the turbine engine **710**. As illustrated, the first angle **1311** can be a non-zero angle, however any angle greater than zero is contemplated.

(151) A second angle **1313** can be defined as the angle between the centerline **842** of the fuel cavity **810** and a fuel orifice centerline **1361** of at least one of the fuel orifices of the second set of fuel orifices **1312***b*. As illustrated, the second angle **1313** can be a non-zero angle, however any angle greater than zero is contemplated.

(152) It is contemplated that the first angle **1311** can be equal to or between 30 degrees to 150

degrees. It is further contemplated that the second angle **1313** can be equal to or between 30 degrees to 150 degrees.

(153) FIG. **26** illustrates another cross section of a portion of a fuel-air mixing assembly **1400**. The fuel-air mixing assembly **1400** is similar to the fuel-air mixing assembly **700**, **800**, **1100**, **1200**, **1300** therefore, like parts will be identified with like numerals increased by 100 with it being understood that the description of the like parts of the fuel-air mixing assembly **700**, **800**, **1100**, **1200**, **1300** applies to the fuel-air mixing assembly **1400**, unless otherwise noted. The fuel-air mixing assembly **1400** includes at least the outer wall **802** with the plurality of apertures **808**, the center body **804**, the air passage **806**, a fuel cavity **1410**, and a plurality of fuel orifices **1412**.

(154) A reducing cross-section area portion or reduction portion **1401** can be formed with or coupled to the outer wall inner surface **820** and include a sloped or angled portion **1405** that decreases the diameter of the air passage **806**. Axially downstream or upstream of the angled portion **1405** can be a constant area portion **1462**, where an air passage diameter **1436** remains constant. While illustrated as a portion of the outer wall inner surface **820**, it is contemplated that the converging, sloped, or angled portion **1405** can extend axially past the second end **830**. It is also contemplated that the downstream end of the angled portion **1405** can be within a distance of the fuel outlet **1454** that is 10% or less of the diameter of the center body **804** at the second end **830**.

(155) It is contemplated that the center body **804**, as illustrated, can have a cylindrical section **1403**, where a diameter **1471** of the center body **804** doesn't change by more than 5%. That is, in the cylindrical section **1403**, the center body **804** is generally a hollow cylindrical shape. The cylindrical section **1403** of the center body **804** can axially overlap the angled portion **1405** the reduction portion **1401**.

(156) The fuel cavity **1410** can be defined by the center body **804**. That is, the fuel cavity **1410** is the hollow center of the center body **804**. A fuel inlet **1452** allows fuel from the fuel cavity **1410** to enter the plurality of fuel orifices **1412**. A fuel outlet **1454** fluidly couples the plurality of fuel orifices **1412** to the air passage **806**. That is, the fuel cavity **1410** is fluidly coupled to the air passage **806** via the plurality of fuel orifices **1412**. Optionally, the fuel cavity **1410** can include channels that fluidly couple the fuel cavity **1410** to the plurality of fuel orifices **1412**.

(157) A fuel orifice centerline **1456** can axially align with the angled portion **1405**, as illustrated. However, it is contemplated that the fuel orifice centerline **1456** can also axially align with the constant area portion **1462**. A fuel orifice distance **1460** can be measured from the fuel orifice centerline **1456** to the second end **830** of the center body **804**. The fuel orifice distance **1460** can be between or equal to 0.0 centimeters and 2.0 centimeters. It is contemplated that the fuel orifice distance **1460** can be between or equal to 0%-50% of the center body length. Additionally, or alternatively, the fuel orifice distance **1460** can be between or equal to 0% to 100% of the diameter of the center body **804** at the second end **830**.

(158) Further aspects of the present disclosure are provided by the subject matter of the following clauses.

(159) A gas turbine engine includes a hydrogen fuel delivery assembly configured to deliver a hydrogen fuel flow, a compressor section configured to compress air flowing therethrough to provide a compressed air flow, and a combustor including a combustion chamber having a burner length L and a burner dome height H, the combustion chamber configured to combust a mixture of the hydrogen fuel flow and the compressed air flow, and the combustion chamber being characterized by a combustor size rating between one inch and seven inches.

(160) The gas turbine engine of the preceding clause, wherein the combustor further includes an outer liner and an inner liner, the combustion chamber having a forward end and being defined between the outer liner and the inner liner, each of the outer liner and the inner liner having an inner surface, and wherein H is the maximum height between the inner surface of the outer liner and the inner surface of the inner liner at the forward end of the combustion chamber.

(161) The gas turbine engine of any preceding clause, wherein the combustor size rating is between

two inches and seven inches.
(162) The gas turbine engine of any preceding clause, wherein the combustor size rating is between two inches and six inches.
(163) The gas turbine engine of any preceding clause, wherein the combustor size rating is between three inches and six inches.
(164) The gas turbine engine of any preceding clause, wherein the burner length is between two inches and six inches.
(165) The gas turbine engine of any preceding clause, wherein the burner length is between two inches and three inches.
(166) The gas turbine engine of any preceding clause, wherein the burner length is between two and one half inches and three and one half inches.
(167) The gas turbine engine of any preceding clause, wherein the burner dome height is between two and one half inches and six inches.
(168) The gas turbine engine of any preceding clause, wherein the burner dome height is between two and one half inches and five inches.
(169) The gas turbine engine of any preceding clause, wherein the burner dome height is between four inches and five inches.
(170) The gas turbine engine of any preceding clause, wherein the burner length, squared, is between six square inches and thirty-five square inches.
(171) The gas turbine engine of any preceding clause, wherein the burner length, squared, is between six square inches and twenty square inches.
(172) The gas turbine engine of any preceding clause, wherein the burner length, squared, is between six square inches and twelve square inches.
(173) The gas turbine engine of any preceding clause, wherein the burner length, squared, is between eight square inches and twelve square inches.
(174) The gas turbine engine of any preceding clause, wherein no diluent is added to the combustion chamber.
(175) The gas turbine engine of any preceding clause, wherein the combustor size rating is defined by a relationship of the burner length, squared, and the burner dome height.
(176) The gas turbine engine of any preceding clause, further comprising a turbine nozzle downstream of the combustion chamber, wherein L is the distance between a plane orthogonal to a forward line at which the burner dome height is measured and a leading edge of the turbine nozzle.
(177) The gas turbine engine of any preceding clause, further comprising one or more rotating blades.
(178) The gas turbine engine of any preceding clause, further comprising one or more stationary vanes, wherein the one or more stationary vanes are positioned forward of the rotating blades.
(179) The gas turbine engine of any preceding clause, further comprising one or more stationary vanes, wherein the one or more stationary vanes are positioned aft of the rotating blades.
(180) The gas turbine engine of any preceding clause, further comprising a first set of one or more rotating blades and a second set of rotating blades, the first set of rotating blades and the second set of rotating blades being configured to operate in a counter-rotating fashion.
(181) The gas turbine engine of any preceding clause, further comprising a plurality of fan blades located forward of the combustor in a puller configuration.
(182) The gas turbine engine of any preceding clause, further comprising a plurality of fan blades, the combustor located forward of the plurality of fan blades in a pusher configuration.
(183) The gas turbine engine of any preceding clause, wherein the gas turbine engine is one of a turbofan engine, an unducted single fan engine, a turbojet engine, a turboshaft engine, or a turboprop engine.
(184) The gas turbine engine of any preceding clause, wherein the gas turbine engine is a turbofan engine comprising an outer nacelle that houses the compressor section, the combustor, and a

plurality of fan blades.
(185) The gas turbine engine of any preceding clause, wherein the gas turbine engine is an unducted single fan engine comprising a spinner coupled to a nacelle, the nacelle housing the compressor section and the combustor, a plurality of outlet guide vanes coupled to an outer surface of the nacelle, and a plurality of fan blades coupled to the spinner and rotatable therewith.
(186) The gas turbine engine of any preceding clause, wherein the gas turbine engine is a turbojet engine comprising an outer nacelle that houses the compressor section and the combustor, the turbojet engine not including a fan with bypass duct.
(187) The gas turbine engine of any preceding clause, further including a hydrogen fuel tank for holding the hydrogen fuel in a liquid phase, the hydrogen fuel delivery assembly being connected to the hydrogen fuel tank, and a vaporizer in communication with the hydrogen fuel delivery assembly for heating the hydrogen fuel in the liquid phase to at least one of a gaseous phase and a supercritical phase, the vaporizer being located between the hydrogen fuel tank and the combustor.
(188) An aircraft including a fuselage, a wing connected to the fuselage, and the gas turbine engine of any preceding clause.
(189) The aircraft of the preceding clause, wherein the hydrogen fuel tank is positioned at least partially within at least one of the fuselage and the wing, and wherein the vaporizer is positioned at least partially within at least one of the fuselage, the wing, and the gas turbine engine.
(190) A gas turbine engine including a hydrogen fuel delivery assembly configured to deliver a hydrogen fuel flow, a compressor section configured to compress air flowing therethrough to provide a compressed air flow, and a combustor including a combustion chamber characterized by a combustor size rating between one inch and seven inches at a core air flow parameter between two and one half kN and sixty kN, wherein the combustor size rating is a function of the core air flow parameter.
(191) The gas turbine engine of any preceding clause, wherein the core air flow parameter is a relationship between the thrust and bypass ratio.
(192) The gas turbine engine of any preceding clause, wherein the combustor size rating is between two inches and three and one quarter inches at a core air flow parameter between two and one half kN and fifty kN.
(193) The gas turbine engine of any preceding clause, wherein the combustor size rating is based on a thrust of the gas turbine engine.
(194) The gas turbine engine of any preceding clause, wherein the thrust is between sixty kN and five hundred kN.
(195) The gas turbine engine of any preceding clause, wherein the combustor size rating is defined by a relationship of the burner length, squared, and the burner dome height.
(196) The gas turbine engine of any preceding clause, further comprising a turbine nozzle downstream of the combustion chamber, wherein the burner length is the distance between a plane orthogonal to a forward line at which the burner dome height is measured and a leading edge of the turbine nozzle.
(197) The gas turbine engine of any preceding clause, wherein the burner length, squared, is between six square inches and thirty-five square inches.
(198) The gas turbine engine of any preceding clause, wherein the combustor further includes an outer liner and an inner liner, the combustion chamber having a forward end and being defined between the outer liner and the inner liner, each of the outer liner and the inner liner having an inner surface, and wherein the burner dome height is the maximum height between the inner surface of the outer liner and the inner surface of the inner liner at the forward end of the combustion chamber.
(199) The gas turbine engine of any preceding clause, wherein the combustor size rating is between two inches and seven inches.
(200) The gas turbine engine of any preceding clause, wherein the combustor size rating is between

two inches and six inches.
(201) The gas turbine engine of any preceding clause, wherein the combustor size rating is between three inches and six inches.
(202) The gas turbine engine of any preceding clause, wherein the burner length is between two inches and six inches.
(203) The gas turbine engine of any preceding clause, wherein the burner length is between two inches and three inches.
(204) The gas turbine engine of any preceding clause, wherein the burner length is between two and one half inches and three and one half inches.
(205) The gas turbine engine of any preceding clause, wherein the burner dome height is between two and one half inches and six inches.
(206) The gas turbine engine of any preceding clause, wherein the burner dome height is between two and one half inches and five inches.
(207) The gas turbine engine of any preceding clause, wherein the burner dome height is between four inches and five inches.
(208) The gas turbine engine of any preceding clause, wherein the burner length, squared, is between six square inches and twenty square inches.
(209) The gas turbine engine of any preceding clause, wherein the burner length, squared, is between six square inches and twelve square inches.
(210) The gas turbine engine of any preceding clause, wherein the burner length, squared, is between eight square inches and twelve square inches.
(211) The gas turbine engine of any preceding clause, wherein no diluent is added to the combustion chamber.
(212) The gas turbine engine of any preceding clause, further comprising one or more rotating blades.
(213) The gas turbine engine of any preceding clause, further comprising one or more stationary vanes, wherein the one or more stationary vanes are positioned forward of the rotating blades.
(214) The gas turbine engine of any preceding clause, further comprising one or more stationary vanes, wherein the one or more stationary vanes are positioned aft of the rotating blades.
(215) The gas turbine engine of any preceding clause, further comprising a first set of one or more rotating blades and a second set of rotating blades, the first set of rotating blades and the second set of rotating blades being configured to operate in a counter-rotating fashion.
(216) The gas turbine engine of any preceding clause, further comprising a plurality of fan blades located forward of the combustor in a puller configuration.
(217) The gas turbine engine of any preceding clause, further comprising a plurality of fan blades, the combustor located forward of the plurality of fan blades in a pusher configuration.
(218) The gas turbine engine of any preceding clause, wherein the gas turbine engine is one of a turbofan engine, an unducted single fan engine, a turbojet engine, a turboshaft engine, or a turboprop engine.
(219) The gas turbine engine of any preceding clause, wherein the gas turbine engine is a turbofan engine comprising an outer nacelle that houses the compressor section, the combustor, and a plurality of fan blades.
(220) The gas turbine engine of any preceding clause, wherein the gas turbine engine is an unducted single fan engine comprising a spinner coupled to a nacelle, the nacelle housing the compressor section and the combustor, a plurality of outlet guide vanes coupled to an outer surface of the nacelle, and a plurality of fan blades coupled to the spinner and rotatable therewith.
(221) The gas turbine engine of any preceding clause, wherein the gas turbine engine is a turbojet engine comprising an outer nacelle that houses the compressor section and the combustor, the turbojet engine not including a fan with bypass duct.
(222) The gas turbine engine of any preceding clause, further including a hydrogen fuel tank for

holding the hydrogen fuel in a liquid phase, the hydrogen fuel delivery assembly being connected to the hydrogen fuel tank, and a vaporizer in communication with the hydrogen fuel delivery assembly for heating the hydrogen fuel in the liquid phase to at least one of a gaseous phase and a supercritical phase, the vaporizer being located between the hydrogen fuel tank and the combustor.
(223) An aircraft including a fuselage, a wing connected to the fuselage, and the gas turbine engine of any preceding clause.
(224) The aircraft of the preceding clause, wherein the hydrogen fuel tank is positioned at least partially within at least one of the fuselage and the wing, and wherein the vaporizer is positioned at least partially within at least one of the fuselage, the wing, and the gas turbine engine.
(225) A gas turbine engine includes a hydrogen fuel delivery assembly configured to deliver a hydrogen fuel flow, a compressor section configured to compress air flowing therethrough to provide a compressed air flow, and a combustion section comprising: at least one fuel-air mixing assembly comprising a center body, an outer wall spaced from and circumscribing the center body, an annular flow passage defined between the outer wall and the center body, and having an inlet and an outlet, and at least one fuel orifice having a fuel outlet opening into the annular flow passage, and a combustor defining a combustion chamber, characterized by a combustor size rating between one inch and seven inches at a core air flow parameter between two and one half kN and sixty kN, wherein the combustor size rating is a function of the core air flow parameter, and wherein the combustor size rating is defined by a relationship of the burner length, squared, and the burner dome height and wherein the core air flow parameter is a relationship between the thrust and bypass ratio.
(226) The turbine engine of any of the proceeding clauses wherein the center body extends axially from a fore end to an aft end and the inlet is at the fore end and the outlet is at the aft end.
(227) The turbine engine of any of the proceeding clauses wherein the annular flow passage includes a constant cross-sectional area portion along a predetermined portion of the center body and terminating at the aft end.
(228) The turbine engine of any of the proceeding clauses, wherein the fuel outlet opening is at a predetermined distance from the aft end of the center body.
(229) The turbine engine of any of the preceding clauses wherein the predetermined distance is 0% to 50% of a center body length.
(230) The turbine engine of any of the preceding clauses wherein the predetermined distance is less than 25% of the center body length.
(231) The turbine engine of any of the preceding clauses wherein the fuel outlet opens at the constant cross-sectional area portion.
(232) The turbine engine of any of the preceding clauses wherein the predetermined distance is between 0.0 to 2.0 centimeters from the aft end.
(233) The turbine engine of any of the preceding clauses wherein the fuel outlet opens at the constant cross-sectional area portion.
(234) The turbine engine of any of the preceding clauses wherein the center body comprises a fuel cavity and the at least one fuel orifice has a fuel inlet fluidly coupled to the fuel cavity.
(235) The turbine engine of any of the preceding clauses wherein the at least one fuel orifice extends through the center body.
(236) The turbine engine of any of the preceding clauses wherein the fuel cavity comprises a channel, extending in the aft-to-fore direction, and the fuel inlet is fluidly coupled to the channel.
(237) The turbine engine of any of the preceding clauses where the channel extends an axial distance between 2% to 50% of the center body length.
(238) The turbine engine of any of the preceding clauses wherein the at least one fuel orifice extends through the outer wall.
(239) The turbine engine of any of the preceding clauses wherein the at least one fuel orifice comprises at least a first set of fuel orifices axially or radially spaced from a second set of fuel

orifices.
(240) The turbine engine of any of the preceding clauses wherein the second set of fuel orifices is circumferentially offset from the first set of fuel orifices.
(241) The turbine engine of any of the preceding clauses wherein at least some of the fuel orifices, of at least one of the first set of fuel orifices or the second set of fuel orifices, are radially angled or axially angled relative to the center body axis.
(242) The turbine engine of any of the preceding clauses wherein the annular flow passage comprises a reducing cross-sectional area portion located upstream of the constant cross-sectional area portion.
(243) The turbine engine of any of the preceding clauses wherein the reducing cross-sectional area portion terminates at the beginning of the constant cross-sectional area portion.
(244) The gas turbine engine of the preceding clause, wherein the combustor further includes an outer liner and an inner liner, the combustion chamber having a forward end and being defined between the outer liner and the inner liner, each of the outer liner and the inner liner having an inner surface, and wherein H is the maximum height between the inner surface of the outer liner and the inner surface of the inner liner at the forward end of the combustion chamber.
(245) The gas turbine engine of any preceding clause, wherein the combustor size rating is between two inches and seven inches.
(246) The gas turbine engine of any preceding clause, wherein the combustor size rating is between two inches and six inches.
(247) The gas turbine engine of any preceding clause, wherein the combustor size rating is between three inches and six inches.
(248) The gas turbine engine of any preceding clause, wherein the burner length is between two inches and six inches.
(249) The gas turbine engine of any preceding clause, wherein the burner length is between two inches and three inches.
(250) The gas turbine engine of any preceding clause, wherein the burner length is between two and one half inches and three and one half inches.
(251) The gas turbine engine of any preceding clause, wherein the burner dome height is between two and one half inches and six inches.
(252) The gas turbine engine of any preceding clause, wherein the burner dome height is between two and one half inches and five inches.
(253) The gas turbine engine of any preceding clause, wherein the burner dome height is between four inches and five inches.
(254) The gas turbine engine of any preceding clause, wherein the burner length, squared, is between six square inches and thirty-five square inches.
(255) The gas turbine engine of any preceding clause, wherein the burner length, squared, is between six square inches and twenty square inches.
(256) The gas turbine engine of any preceding clause, wherein the burner length, squared, is between six square inches and twelve square inches.
(257) The gas turbine engine of any preceding clause, wherein the burner length, squared, is between eight square inches and twelve square inches.
(258) The gas turbine engine of any preceding clause, wherein no diluent is added to the combustion chamber.
(259) The gas turbine engine of any preceding clause, further comprising a turbine nozzle downstream of the combustion chamber, wherein L is the distance between a plane orthogonal to a forward line at which the burner dome height is measured and a leading edge of the turbine nozzle.
(260) The gas turbine engine of any preceding clause, further comprising one or more rotating blades.
(261) The gas turbine engine of any preceding clause, further comprising one or more stationary

vanes, wherein the one or more stationary vanes are positioned forward of the rotating blades.
(262) The gas turbine engine of any preceding clause, further comprising one or more stationary vanes, wherein the one or more stationary vanes are positioned aft of the rotating blades.
(263) The gas turbine engine of any preceding clause, further comprising a first set of one or more rotating blades and a second set of rotating blades, the first set of rotating blades and the second set of rotating blades being configured to operate in a counter-rotating fashion.
(264) The gas turbine engine of any preceding clause, further comprising a plurality of fan blades located forward of the combustor in a puller configuration.
(265) The gas turbine engine of any preceding clause, further comprising a plurality of fan blades, the combustor located forward of the plurality of fan blades in a pusher configuration.
(266) The gas turbine engine of any preceding clause, wherein the gas turbine engine is one of a turbofan engine, an unducted single fan engine, a turbojet engine, a turboshaft engine, or a turboprop engine.
(267) The gas turbine engine of any preceding clause, wherein the gas turbine engine is a turbofan engine comprising an outer nacelle that houses the compressor section, the combustor, and a plurality of fan blades.
(268) The gas turbine engine of any preceding clause, wherein the gas turbine engine is an unducted single fan engine comprising a spinner coupled to a nacelle, the nacelle housing the compressor section and the combustor, a plurality of outlet guide vanes coupled to an outer surface of the nacelle, and a plurality of fan blades coupled to the spinner and rotatable therewith.
(269) The gas turbine engine of any preceding clause, wherein the gas turbine engine is a turbojet engine comprising an outer nacelle that houses the compressor section and the combustor, the turbojet engine not including a fan with bypass duct.
(270) The gas turbine engine of any preceding clause, further including a hydrogen fuel tank for holding the hydrogen fuel in a liquid phase, the hydrogen fuel delivery assembly being connected to the hydrogen fuel tank, and a vaporizer in communication with the hydrogen fuel delivery assembly for heating the hydrogen fuel in the liquid phase to at least one of a gaseous phase and a supercritical phase, the vaporizer being located between the hydrogen fuel tank and the combustor.
(271) An aircraft including a fuselage, a wing connected to the fuselage, and the gas turbine engine of any preceding clause.
(272) The aircraft of the preceding clause, wherein the hydrogen fuel tank is positioned at least partially within at least one of the fuselage and the wing, and wherein the vaporizer is positioned at least partially within at least one of the fuselage, the wing, and the gas turbine engine.
(273) A gas turbine engine includes a hydrogen fuel delivery assembly configured to deliver a hydrogen fuel flow, a compressor section configured to compress air flowing therethrough to provide a compressed air flow, and a combustor including a combustion chamber characterized by a combustor size rating between one inch and seven inches at a core air flow parameter between two and one half kN and sixty kN, wherein the combustor size rating is a function of the core air flow parameter, and wherein the combustor size rating is defined by a relationship of the burner length, squared, and the burner dome height and wherein the core air flow parameter is a relationship between the thrust and bypass ratio.
(274) Although the foregoing description is directed to the preferred embodiments, it is noted that other variations and modifications will be apparent to those skilled in the art, and may be made without departing from the spirit or scope of the disclosure Moreover, features described in connection with one embodiment may be used in conjunction with other embodiments, even if not explicitly stated above.

Claims

1. A gas turbine engine comprising: a hydrogen fuel delivery assembly configured to deliver a hydrogen fuel flow; a compressor section configured to compress air flowing therethrough to provide a compressed air flow; and a combustion section comprising: at least one fuel-air mixing assembly comprising a center body, an outer wall spaced from and circumscribing the center body, an annular flow passage defined between the outer wall and the center body, and having an inlet and an outlet, and at least one fuel orifice having a fuel outlet opening into the annular flow passage, and a combustor configured to operate without diluent, the combustor including an inner liner, an outer liner, and a combustion chamber, the combustion chamber characterized by a combustor size rating between one inch and seven inches at a core air flow parameter between two and one half kN and sixty kN, wherein the combustor size rating is a function of the core air flow parameter, and wherein the combustor size rating is defined by: $\frac{L^2}{H}$ wherein His a maximum height of the combustion chamber measured by a forward line extending from an inner surface of the outer liner to an inner surface of the inner liner and L is a length of the combustion chamber measured from a midpoint of the forward line to a midpoint of an aft line, the aft line extending from the inner surface of the inner liner to the inner surface of the outer liner at a leading edge of a turbine nozzle, and, wherein the core air flow parameter is defined by: $\frac{\text{Thrust}}{\text{BypassRatio}}$.
2. The gas turbine engine of claim 1, wherein the center body extends axially from a fore end to an aft end and the inlet is at the fore end and the outlet is at the aft end.
3. The gas turbine engine of claim 2, wherein the annular flow passage includes a constant cross-sectional area portion along a predetermined portion of the center body and terminating at the aft end.
4. The gas turbine engine of claim 3, wherein the fuel outlet opening is at a predetermined distance from the aft end of the center body.
5. The gas turbine engine of claim 4 wherein the predetermined distance is less than or equal to 50% of a center body length.
6. The gas turbine engine of claim 4 wherein the predetermined distance is less than 25% of a center body length.
7. The gas turbine engine of claim 6 wherein the fuel outlet opens at the constant cross-sectional area portion.
8. The gas turbine engine of claim 3 wherein the predetermined distance is between 0.0 to 2.0 centimeters from the aft end.
9. The gas turbine engine of claim 3 wherein the fuel outlet opens at the constant cross-sectional area portion.
10. The gas turbine engine of claim 1 wherein the center body comprises a fuel cavity and the at least one fuel orifice has a fuel inlet fluidly coupled to the fuel cavity and wherein the at least one fuel orifice extends through the center body.
11. The gas turbine engine of claim 10 wherein the fuel cavity comprises a channel, extending in an aft-to-fore direction, and the fuel inlet is fluidly coupled to the channel.
12. The gas turbine engine of claim 11 where the channel extends an axial distance between 2% to 50% of a center body length.
13. The gas turbine engine of claim 1 wherein the at least one fuel orifice extends through the outer wall.
14. The gas turbine engine of claim 1 wherein the at least one fuel orifice comprises at least a first set of fuel orifices axially or radially spaced from a second set of fuel orifices.
15. The gas turbine engine of claim 14 wherein the second set of fuel orifices is circumferentially offset from the first set of fuel orifices or at least some of the fuel orifices, of at least one of the first set of fuel orifices or the second set of fuel orifices, are radially angled or axially angled relative to a center body axis.
16. The gas turbine engine of claim 1 wherein the annular flow passage includes a constant cross-

sectional area portion along a predetermined portion of the center body and terminating at an aft end and wherein the annular flow passage comprises a reducing cross-sectional area portion located upstream of the constant cross-sectional area portion.

17. The gas turbine engine of claim 16 wherein the reducing cross-sectional area portion terminates at the beginning of the constant cross-sectional area portion.

18. The gas turbine engine of claim 1, wherein the combustor size rating is between two inches and three and one quarter inches at a core air flow parameter between two and one half kN and fifty kN.

19. The gas turbine engine of claim 1, wherein the combustor size rating is based on a thrust of the gas turbine engine.